

Segmenting a Survey Population Without Overstating What the Design Supports

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1 Purpose and scope

This document is both the methodology of a design-based latent class analysis (LCA) workflow and a worked application of it. Every estimator is stated where it is first used and immediately applied, so a reader can follow the reasoning and the results together. The application is the institutional trust battery from the 2023 LAPOP AmericasBarometer survey of Ecuador; the survey is an example, and the methods are the subject.

Throughout, a latent class is called a **segment**. This is a word-choice preference for applied audiences; the statistical object is unchanged.

The workflow's organizing commitment is that **one replicate design generates every standard error in the analysis**. The same stratified jackknife replicate weights that produce the interval on a conditional response probability produce the interval on a segment prevalence within a demographic group. Nothing is computed under an independence assumption at any stage.

The closest existing software is the `baysc` package [1], [2], which fits a weighted objective within a Bayesian pseudo-posterior with a post-hoc variance adjustment. It differs in three respects: it selects the number of segments automatically through an overfitted mixture, it obtains uncertainty from an adjusted posterior rather than replicate weights, and being Bayesian it propagates classification uncertainty through the posterior draws, which makes the attenuation discussed in Section 15 a non-issue in that framework. The two are complements. This workflow is the frequentist, replicate-variance counterpart, and it has base survey dependencies only, which matters in analytic environments restricted to CRAN.

Three files feed this report. `survey_data_read.R` reads the file and cleans the design columns and demographics, which both arms share, so the recode block exists in exactly one place. `survey_lca_config.R` names the items and the settings for this arm. `source_code.R` is the estimation engine and is never edited per dataset. Section 18 documents what a port requires.

2 Where this sits in the Total Survey Error framework

The Total Survey Error (TSE) framework enumerates the distinct sources of error that arise as a survey is conducted, organized around the survey life cycle and split into two arms [3], [4]. The **representation** arm runs from the target population through coverage error to the sampling frame, sampling error to the sample, nonresponse error to the respondents, and adjustment error to the survey statistic. The **measurement** arm runs from the construct through validity error to the measurement, measurement error

to the response, and processing error to the edited data. Errors attach to *transitions*, not to the survey as a whole, which is why several are properties of an estimate rather than of the survey.

Locating this workflow in that framework clarifies what it does and does not accomplish.

- **Sampling error is what this workflow is chiefly about.** Sampling error is the one representation error that is intended and controllable by design, and under a sound design it is variance rather than bias. The difficulty is that estimating it correctly requires using the realized design, and a recognized weakness of practice is that many organizations report only sampling error while computing it as though the sample were simple and random. A latent class model fit to a stratified, clustered sample and analyzed with default software produces margins of error that are too narrow, and the understatement can be large. Section 9 addresses this directly.
- **Adjustment error is engaged but not removed.** Post-survey weighting reduces nonresponse and noncoverage bias at the cost of increased sampling variance, and that increase is itself a source of error. This workflow carries the released weights into every step of estimation so that segments describe the population, and reports the unequal weighting effect so the variance cost is visible rather than implicit.
- **Nonresponse error is addressed at the item level only.** Respondents who skipped some items are scored from the items they answered rather than discarded, and all domain estimation runs on that larger frame, because item nonresponse on attitude batteries is not random (Section 14). Unit nonresponse is handled by the data provider's weights and is not modeled here.
- **Measurement error is the reason to use a latent class model at all.** LCA is used in survey research precisely because it accommodates error-prone categorical indicators: the latent segment is the construct, the items are its fallible measurements, and the conditional response probabilities are estimates of how the measurement behaves within each segment. The model does not remove measurement error; it makes it a parameter.
- **Processing error is introduced by this workflow and must be acknowledged.** Two steps add error after the response is recorded. Assigning each respondent to a single most likely segment introduces classification error, which attenuates associations between segments and covariates (Section 15). Drafting segment names with a language model is a processing step whose output is a label, reviewed and owned by the analyst, that never enters a computation (Section 12).
- **Coverage error and validity are out of scope.** They are properties of the frame and the questionnaire, fixed before any analysis begins.

Two caveats about the framework itself are worth carrying forward. TSE centers on accuracy and is largely silent on the other dimensions of data quality: relevance, timeliness, accessibility, interpretability, and coherence [5]. And its tree structure can imply that error sources are independent when they are frequently correlated; the classification error introduced in Section 15, for example, is larger for respondents who answered fewer items, which links a processing error to an item nonresponse error.

3 Choosing between a latent class model and a factor model

Both models take the same battery of answers and look for something underneath that the individual questions only measure imperfectly. They differ in what they assume that something is. A latent class model assumes it is a small number of distinct kinds of respondent, and asks which kind each person most resembles. A factor model assumes it is a single continuum running from low to high, and asks where on it each person sits. Neither assumption is more correct in general; the battery decides.

The practical question is whether people differ in *how much* or in *which*. If respondents mostly line up from those who answer low on everything to those who answer high on everything, a continuum describes them and the classes are just cut points on it. If instead a substantial group trusts some institutions while distrusting others, and a different group reverses that pattern, then no single number can represent both and the class model is recovering something real.

The class model has a communication advantage that is worth weighing. Its output is a share of a group, and shares are what most readers already know how to interpret: saying that thirty-one per cent of rural respondents fall in the distrustful segment needs no explanation. A factor model produces a position on a scale with no natural units, which has to be translated before anyone outside the analysis can use it. That is a reason to prefer classes when the substantive difference between the two models is small, and it is not a reason to prefer them when it is large.

Both models rest on the same central assumption, called local independence: that once you know a person's class, or their position on the factor, their answers to the individual questions carry no further relationship to each other. This is what allows either model to be summarised so compactly, and it is the first thing to fail when the battery contains two questions that are really asking the same thing. The report checks it directly and reports where it strains.

The assumptions that differ are these. The class model assumes classes are genuinely discrete rather than convenient cut points on a continuum, and it does not assume the response categories are ordered, which is why it can pick up a group that avoids the middle of a scale or clusters on one particular answer. The factor model assumes the relationship between each question and the underlying continuum is monotonic, so that more of the trait always means a higher answer, and in exchange it estimates far fewer quantities and is much more stable at small sample sizes. An item that violates monotonicity can separate classes strongly while contributing almost nothing to a factor, and the reverse also happens. Where the two disagree about an item, that disagreement is information about the battery rather than a problem to be resolved by picking one model.

4 Data and preparation

The 2023 AmericasBarometer round in Ecuador interviewed voting-age adults under a stratified, clustered design with released base weights [6]. We analyze an eleven-item battery on the material circumstances of the household and the respondent's own situation: durable goods, running short of food or water, perceived change in personal finances, neighbourhood safety and crime victimisation, receipt of government assistance and of a conditional cash transfer, and intention to emigrate. The items carry two, three and four response categories and span unrelated domains, which is the case a class model handles natively and a factor model cannot.

Preparation happens in `survey_data_read.R` and `survey_lca_config.R`, and follows three rules. Non-response codes are recoded to missing, with the source file read so those codes are visible rather than silently converted on import. Items are recoded to consecutive integers over their observed categories on *every* row of the file, so the estimation frame and the prediction frame of Section 14 share one coding by construction. Demographic recodes are written as one exhaustive `recode_values` per variable with `unmatched = "error"`, so the render halts if any source label falls through, since a pattern-matching scheme can silently misclassify or delete respondents.

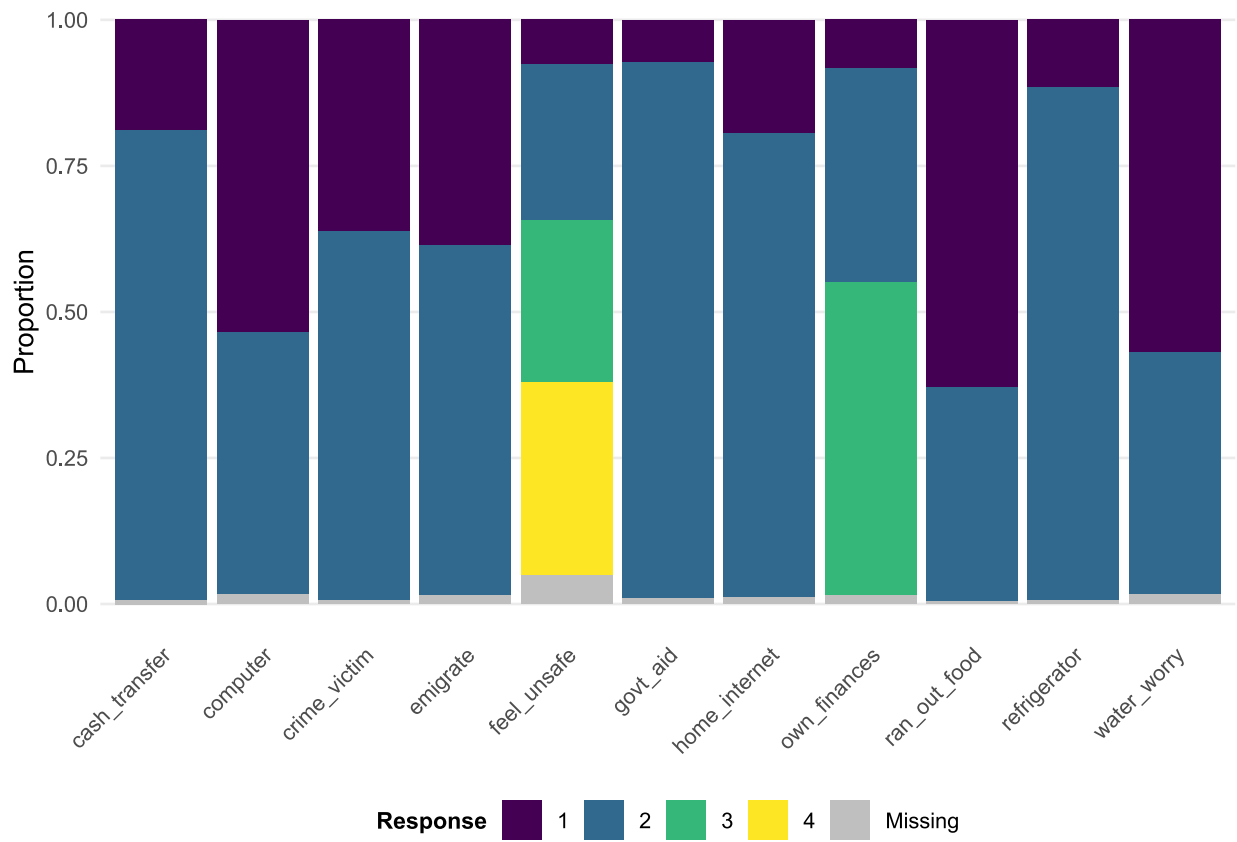


Figure 1: All respondents, all items, with missingness shown explicitly in grey. The analysis rule removes the grey.

1445 of 1604 respondents enter estimation.

Table 1: Recode audit: what every source label became.

Variable	Source label	Recoded to	n
age_cat	28	16-29	70
age_cat	16	16-29	65
age_cat	27	16-29	61
age_cat	44	30-44	59
age_cat	19	16-29	50
age_cat	24	16-29	47
age_cat	20	16-29	46
age_cat	29	16-29	46
age_cat	23	16-29	45
age_cat	45	45-59	45
age_cat	18	16-29	43
age_cat	22	16-29	41
age_cat	25	16-29	41
age_cat	40	30-44	40
age_cat	17	16-29	36
age_cat	26	16-29	35
age_cat	30	30-44	35
age_cat	43	30-44	31

Variable	Source label	Recoded to	n
age_cat	46	45-59	29
age_cat	50	45-59	29
age_cat	21	16-29	27
age_cat	37	30-44	27
age_cat	38	30-44	27
age_cat	42	30-44	27
age_cat	36	30-44	26
age_cat	33	30-44	24
age_cat	34	30-44	24
age_cat	35	30-44	24
age_cat	47	45-59	24
age_cat	55	45-59	22
age_cat	31	30-44	21
age_cat	32	30-44	20
age_cat	41	30-44	20
age_cat	52	45-59	20
age_cat	54	45-59	20
age_cat	58	45-59	20
age_cat	60	60+	20
age_cat	65	60+	20
age_cat	56	45-59	19
age_cat	57	45-59	17
age_cat	62	60+	17
age_cat	49	45-59	16
age_cat	64	60+	16
age_cat	39	30-44	15
age_cat	51	45-59	15
age_cat	53	45-59	15
age_cat	70	60+	14
age_cat	63	60+	13
age_cat	67	60+	13
age_cat	68	60+	13
age_cat	69	60+	12
age_cat	48	45-59	11
age_cat	72	60+	11
age_cat	61	60+	10
age_cat	66	60+	10
age_cat	73	60+	9
age_cat	59	45-59	8
age_cat	71	60+	7
age_cat	76	60+	7
age_cat	74	60+	5
age_cat	75	60+	5
age_cat	77	60+	3
age_cat	78	60+	3
age_cat	80	60+	3
age_cat	81	60+	2
age_cat	79	60+	1
age_cat	82	60+	1
age_cat	83	60+	1
age_cat	84	60+	1
age_cat	85	60+	1
age_cat	86	60+	1
age_cat	88	60+	1
age_cat	89	60+	1
education	Secundaria/bachillerato completo	Secondary	557

Variable	Source label	Recoded to	n
education	Secundaria/bachillerato incompleto	Secondary	241
education	Terciaria o universitaria o superior incompleta	Tertiary	233
education	Primaria/básica completa	Primary	221
education	Terciaria o universitaria o superior completa	Tertiary	214
education	Primaria/básica incompleta	Primary	91
education	NR	NA	26
education	DK	NA	13
education	Ninguna	NA	8
employment	Trabajando?	Employed	586
employment	Está buscando trabajo activamente?	Unemployed	272
employment	Se dedica a los quehaceres de su hogar?	Homemaker	256
employment	Es estudiante?	Student	182
employment	No está trabajando en este momento pero tiene trabajo?	Employed	111
employment	Está jubilado, pensionado o incapacitado permanentemente para trabajar?	Retired	88
employment	No trabaja y no está buscando trabajo?	Not in labor force	76
employment	NR	NA	18
employment	DK	NA	15
sex	Mujer/femenino	Female	799
sex	Hombre/masculino	Male	786
sex	DK	NA	10
sex	NR	NA	8
sex	No se identifica como hombre ni como mujer	NA	1
urban	Urbano	Urban	1073
urban	Rural	Rural	531

Table 2: Item dictionary, with response labels in fitted-category order.

Analysis name	Source variable	Survey question	Response labels
refrigerator	r3	Tiene un refrigerador (nevera) en el hogar	No Sí
computer	r15	Tiene una computadora, computadora portátil, tableta, iPad en el hogar	No Sí
home_internet	r18	Tiene acceso a internet en el hogar, incluyendo teléfono o tableta	No Sí
ran_out_food	fs2	Se ha quedado sin comida en los últimos 3 meses	No Sí
water_worry	ws1	Se ha quedado sin agua potable en los últimos 3 meses	No Sí
own_finances	idio2	Percepción de situación económica personal (En los últimos 12 meses)	Mejor Igual Peor
feel_unsafe	aoj11	Percepción de seguridad en el vecindario	Muy seguro(a) Algo seguro(a) Algo inseguro(a) Muy inseguro(a)
crime_victim	vic1ext	Víctima del crimen en los últimos 12 meses	Sí No
govt_aid	wf1	Asistencia periódica en forma de dinero, alimentos o productos del gobierno	Sí No
cash_transfer	cct1b	Beneficiario del programa de transferencias condicionadas	Sí No
emigrate	q14	Intención de vivir/trabajar en otro país (próximos tres años)	Sí No

```

dat <- cfg$data
w_vec <- dat[[cfg$weight]]
n_obs <- nrow(dat)
scale_ic <- n_obs / sum(w_vec)
cv_w <- sd(w_vec) / mean(w_vec)

glue("Weights: CV = {round(cv_w, 3)}, ",
     "unequal weighting effect = {round(1 + cv_w^2, 2)}.")

```

Weights: CV = 0.055, unequal weighting effect = 1.

The unequal weighting effect $1 + CV^2(w)$ is the multiplicative variance penalty that weight variation imposes even when the weights are unrelated to the survey variable [7]. It is the adjustment-error cost referred to in Section 2. When it is close to 1 the weighted and unweighted fits will differ little, and the comparison in Section 11 should be read accordingly.

5 The latent class model

Let $\mathbf{y}_i = (y_{i1}, \dots, y_{iJ})$ be respondent i 's answers to J categorical items, item j having C_j categories. With K segments, prevalences π_k , and conditional response probabilities $\rho_{jck} = P(y_{ij} = c \mid X_i = k)$ where X_i denotes latent membership,

$$P(\mathbf{y}_i) = \sum_{k=1}^K \pi_k \prod_{j=1}^J \rho_{j y_{ij} k}, \quad \sum_{k=1}^K \pi_k = 1, \quad \sum_{c=1}^{C_j} \rho_{jck} = 1.$$

The product over items is **local independence**: conditional on segment, items are independent [8], [9]. It is the assumption that makes the model parsimonious, that licenses scoring partially observed respondents in Section 14, and that fails first under misspecification, which is why Section 10 probes it. Model dimension is

$$d_K = (K - 1) + K \sum_{j=1}^J (C_j - 1),$$

one free parameter per prevalence beyond the last and $C_j - 1$ free conditional probabilities per item per segment. the model dimension above is what the information criteria in Section 7 penalise. This grows linearly in K with slope $\sum_j (C_j - 1)$, so batteries with many response categories become expensive quickly. The 11 items analysed here carry $\sum_j (C_j - 1) = 14$, giving 59 free parameters at $K = 4$. The factor arm's battery of seven-category items carries $\sum_j (C_j - 1) = 6J$: over the eleven items it analyses that is 66, and 334 free parameters at $K = 5$, which is why the choice of battery and the choice of model are not separable questions. The counts for this battery are computed by `df_k()` rather than written into this sentence, so they cannot drift from the configuration; the figure for the other battery is quoted for eleven items and has to be revisited if that battery changes.

6 Design-weighted estimation

Under a complex design with base weights w_i , estimation maximizes the pseudo-log-likelihood

$$\ell_w(\theta) = \sum_{i=1}^n w_i \log \left(\sum_{k=1}^K \pi_k \prod_{j=1}^J \rho_{j y_{ij} k} \right),$$

the standard device for latent class analysis under complex sampling [10]–[13]. This is a pseudo-likelihood, not a likelihood: it is the sample-based estimate of the census log-likelihood that would be computed if the whole population were observed. Its maximizer is consistent for the census parameter under standard conditions, but it is not the maximizer of a probability model for the observed data, which is why the design must supply the variance rather than the curvature of ℓ_w .

The EM algorithm [14] alternates an expectation step computing posterior membership,

$$p_{ik} = \frac{\pi_k \prod_j \rho_{j y_{ij} k}}{\sum_{l=1}^K \pi_l \prod_j \rho_{j y_{ij} l}},$$

and a maximization step in which weights enter every sufficient statistic,

$$\hat{\pi}_k = \frac{\sum_i w_i p_{ik}}{\sum_i w_i}, \quad \hat{\rho}_{jck} = \frac{\sum_i w_i p_{ik} \mathbf{1}(y_{ij} = c)}{\sum_i w_i p_{ik} \mathbf{1}(y_{ij} \text{ observed})}.$$

Three implementation points in the E-step above and the M-step above are load-bearing rather than incidental. Restricting the $\hat{\rho}$ denominator to respondents who answered item j is what keeps $\sum_c \hat{\rho}_{jck} = 1$ under item nonresponse; with complete cases the restriction is vacuous. The expectation step accumulates $\log \rho$ across items and applies the log-sum-exp identity $\log \sum_k e^{z_k} = z_{\max} + \log \sum_k e^{z_k - z_{\max}}$ before exponentiating, because with many items the raw product underflows to zero and the E-step above becomes 0/0. And a missing item contributes zero on the log scale, which is exactly equivalent to dropping it from the product in the mixture above; this equivalence, not an approximation, is what Section 14 relies on.

Iteration stops when $|\Delta \ell_w| < \varepsilon(|\ell_w| + 1)$, with $\varepsilon = 10^{-8}$ during enumeration and 10^{-10} for the chosen model and every replicate refit. Matching the replicate tolerance to the point-estimate tolerance matters because the jackknife variance is a sum of squared deviations from the full-sample estimate, so systematically under-converged replicates would enter as a bias in every one of those deviations.

Because the pseudo-likelihood of a finite mixture can be multimodal with near-tied optima, each candidate model is fit from `cfg$n_starts` random starts and the reported solution is the best mode found; the claim is stated rather than hidden. Starting values are generated in the main session from a deterministic seed sequence and passed to parallel workers as data, so no worker touches the random number generator and results are identical under sequential and parallel execution. Point estimates are invariant to the scale of the weights.

The likelihood is invariant to permutation of segment labels, so any two fits may agree completely and still not be comparable coordinate by coordinate. We align a fit to a reference by minimizing total squared distance between their response profiles over permutations,

$$\hat{\sigma} = \arg \min_{\sigma \in \mathfrak{S}_K} \sum_{k=1}^K \text{parallel} \rho_{\dots \sigma(k)}^{\text{fit}} - \rho_{\dots k}^{\text{ref}} \text{parallel},$$

solved exactly by the Hungarian algorithm [15], [16] rather than by heuristic matching. Alignment is applied to replicate refits before they enter the variance, to the unweighted benchmark before comparison, and to any comparison across K .

7 Model selection

An information criterion compares a maximized log-likelihood against a penalty in units of $\log(\text{sample size})$, and the two must sit on the same scale. Base weights rarely sum to n . Because ℓ_w is linear in the weights, multiplying it by $n / \sum_i w_i$ is exactly equivalent to refitting with weights rescaled to sum to n : point estimates are untouched and only the criteria change. With $\ell_w^* = \ell_w \cdot n / \sum_i w_i$,

$$\text{AIC} = -2\ell_w^* + 2d_K, \quad \text{BIC} = -2\ell_w^* + d_K \log n.$$

Without the rescaling in the rescaling above, weights summing to substantially more than n inflate the log-likelihood relative to a fixed penalty and BIC leans toward too many segments [17].

We report alongside these the relative entropy of the posterior classification, on the weighted scale,

$$R_E^2 = 1 - \frac{-\sum_i w_i \sum_k p_{ik} \log p_{ik}}{(\sum_i w_i) \log K},$$

which indexes classification sharpness [18]. The numerator of the entropy above is the total weighted posterior uncertainty and the denominator its maximum, attained when every respondent's posterior is uniform over the K segments, so R_E^2 runs from 0 at no classification information to 1 at perfect separation; values near 0.8 and above are conventionally read as good classification. Weighting is not cosmetic here: an unweighted entropy would describe the achieved sample's classification sharpness while every other quantity in the analysis describes the population.

The bootstrap likelihood ratio test, often preferred for class enumeration under simple random sampling [19], is omitted by design. Its parametric bootstrap draws new samples of independent observations from the fitted model, which is not the process that generated a stratified clustered sample. Mplus disables the test under complex-design analysis for the same reason, and nothing is gained by reporting a p-value whose null distribution is known to be wrong.

The number of segments is never selected automatically. The analyst reviews the criteria, the entropy, the profiles, and the diagnostics of Section 10, then fixes K in configuration, where the decision is recorded and auditable.

```
enum <- future_map(cfg$K_range, function(K) {
  fit <- fit_lca(dat, w_vec, cats, cfg$items, K, start_seeds(cfg, K))
  ll_n <- fit$ll * scale_ic
  d <- df_k(K, cats)
  tibble(K = K, logLik = ll_n, df = d,
    AIC = -2 * ll_n + 2 * d,
    BIC = -2 * ll_n + d * log(n_obs),
    entropy = entropy_R2(fit$post, w_vec, K),
    converged = fit$converged)
}, .options = furrr_options(seed = NULL,
  packages = c("matrixStats", "clue", "purrr", "dplyr", "tidyr"))) |>
list_rbind()
```

```
enum |>
  mutate(across(c(logLik, AIC, BIC), \(x) round(x, 1)),
    entropy = round(entropy, 3)) |>
  lca_table(align = "r",
    caption = "Weighted fit statistics by number of segments. Nothing is
selected here.")
```

Table 3: Weighted fit statistics by number of segments. Nothing is selected here.

K	logLik	df	AIC	BIC	entropy	converged
2	-9889.0	29	19836.1	19989.1	0.676	TRUE
3	-9774.8	44	19637.5	19869.7	0.669	TRUE
4	-9698.7	59	19515.4	19826.7	0.710	TRUE
5	-9656.0	74	19460.0	19850.5	0.678	TRUE
6	-9627.0	89	19432.0	19901.6	0.655	TRUE
7	-9602.3	104	19412.7	19961.4	0.691	TRUE
8	-9579.4	119	19396.7	20024.5	0.717	TRUE
9	-9560.1	134	19388.1	20095.1	0.768	TRUE
10	-9541.0	149	19380.1	20166.2	0.767	TRUE
11	-9521.3	164	19370.6	20235.9	0.737	TRUE
12	-9507.8	179	19373.5	20317.9	0.756	TRUE

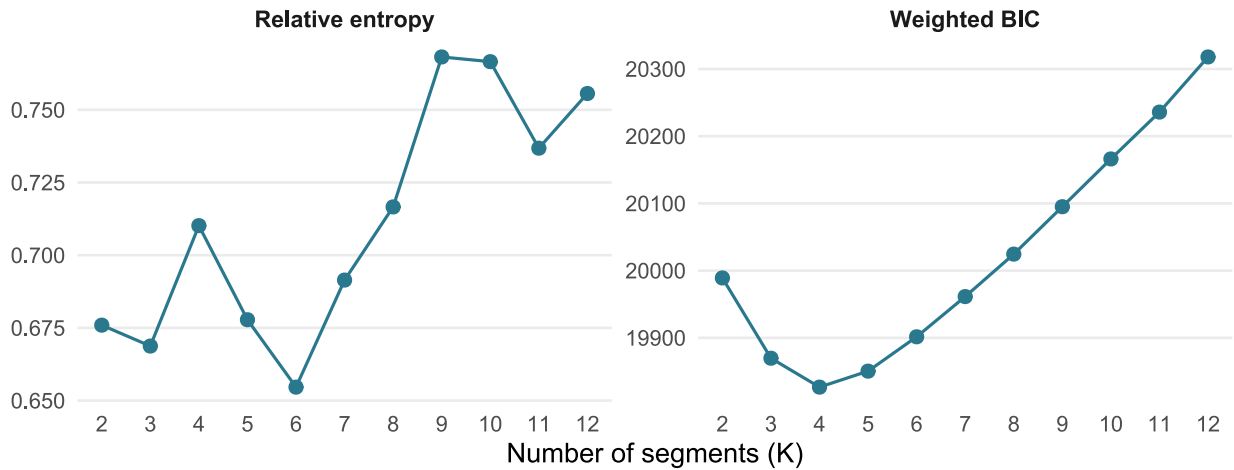


Figure 2: Enumeration diagnostics across K. BIC penalizes model dimension; relative entropy indexes classification sharpness. Neither decides alone.

8 The chosen model

```
if (!K_ready)
  cat("***cfg$K_force` is `NULL`.** Read the evidence above, set the number of",
    "segments in `survey_lca_config.R`, then re-render. Everything below",
    "waits on that.\n")
```

```
fit_w <- fit_lca(dat, w_vec, cats, cfg$items, K_star,
  start_seeds(cfg, K_star), maxit = 5000L, tol = 1e-10)
fit_u <- align_to(fit_lca(dat, rep(1, n_obs), cats, cfg$items, K_star,
```

```

      start_seeds(cfg, K_star), maxit = 5000L, tol = 1e-10),
    fit_w)

inp <- make_inputs(dat, cfg$items, cats)
post_w <- posterior_of(fit_w$pi, fit_w$rho, inp$Y)
colnames(post_w) <- paste0("post_segment", seq_len(K_star))
modal_int <- max.col(post_w, ties.method = "first")

glue(
  "K = {K_star} on {n_obs} cases from {cfg$n_starts} starts. ",
  "Converged: {if (isTRUE(fit_w$converged)) 'yes' else 'NO, inspect before trusting'}.",
  "\n",
  "Weighted log-likelihood (sum-to-n scale): {round(fit_w$ll * scale_ic, 1)}\n",
  "Weighted relative entropy: {round(entropy_R2(post_w, w_vec, K_star), 3)} | ",
  "mean maximum posterior: {round(mean(post_w[cbind(seq_len(n_obs), modal_int)]), 3)}")

```

```

K = 4 on 1445 cases from 200 starts. Converged: yes.
Weighted log-likelihood (sum-to-n scale): -9698.7
Weighted relative entropy: 0.71 | mean maximum posterior: 0.836

```

9 Design-based variance estimation

All uncertainty in this document comes from stratified delete-one-PSU jackknife replication [20]–[22]. With a_h primary sampling units (PSUs, or clusters) in stratum h , the replicate that deletes PSU α of stratum h assigns weight zero to that PSU's respondents, multiplies the weights of the remaining respondents in stratum h by $a_h/(a_h - 1)$, and leaves other strata unchanged. Writing $\hat{\theta}_{(h\alpha)}$ for the estimator recomputed on those weights,

$$\hat{V}(\hat{\theta}) = \sum_{h=1}^H \frac{a_h - 1}{a_h} \sum_{\alpha=1}^{a_h} (\hat{\theta}_{(h\alpha)} - \hat{\theta}) (\hat{\theta}_{(h\alpha)} - \hat{\theta}), \quad \nu = \sum_h a_h - H,$$

where ν is the design degrees of freedom, PSUs minus strata, used for every t reference in this document rather than a normal quantile.

Replication is a choice here, not a necessity: a pseudo-likelihood does admit a sandwich variance [23], and that is what Mplus and Stata compute under survey estimation. We choose replication for three reasons specific to this problem. The parameter vector is moderate here, 59 free parameters at $K = 4$, but grows linearly in K and in the number of response categories, it carries many parameters at or near the boundary, and the linearized sandwich requires inverting an information matrix of that dimension at every stage where a new statistic is wanted. Replication propagates to new statistics without rederivation: once the replicate weights exist, the domain estimates of Section 15 inherit design-based variance by recomputation, where linearization would need a fresh delta-method derivation at each stage. And replication makes the design specification auditable, since a misspecified cluster identifier changes the replicate structure visibly.

Each replicate refits the weighted EM on the replicate weights, warm-started from the full-sample solution and aligned to it by the alignment criterion above. The resulting variance is therefore **within-mode**

by construction: it expresses design variance around the reported solution, not uncertainty about which mode the likelihood surface should have selected.

A stratum containing a single PSU cannot receive the $a_h/(a_h - 1)$ scaling. The `survey.lonely.psu` option governs linearization variance and does not change how replicate weights are constructed, so a singleton stratum would contribute no variance and the resulting standard errors would be anticonservative in exactly the strata where the data are thinnest. The workflow therefore treats singleton strata as a hard stop.

The design is built once, on the whole analysis frame, and is never rebuilt on a case-excluded subset. Rebuilding on a subset is the conditional subpopulation approach: it can drop a PSU that contributed no surviving respondent, which changes a_h , the $a_h/(a_h - 1)$ scaling and the design degrees of freedom, so the resulting interval is no longer the design variance of a domain in the sample that was actually drawn. SURV701 states the rule directly — subset the design, not the data — and names dropped PSUs and singleton computation units as the failure mode. The unconditional approach keeps the replicate structure of the realised sample and restricts the replicate weights to the rows of interest through a keep index; for a ratio estimator, and for the weighted EM, a zero-weight row and an absent row contribute identically, so this is the same computation at the same cost. Earlier versions of this document rebuilt the design on the item-complete frame and again on the scored frame. No PSU was lost on this file — both return 84 replicates and 81 degrees of freedom — so no number here changes, but the construction was wrong and the corrected form is what `build_rep_design()` now does.

One survey option is load-bearing and easy to leave at its default. `as.svrepdesign()` takes `mse` from `getOption("survey.replicates.mse")`, which is `FALSE`, and `svrVar()` then centres the replicate spread on the mean of the replicates. `replicate_variance()` below centres on the full-sample estimate, which is the JK_n formula above. Left at the default, the design-based rows of the domain tables in Section 15 (computed by `svyby`) and the corrected rows (computed by `replicate_variance()`) would come from two different variance estimators, and this document would compare their widths as though they were one. `build_rep_design()` sets `mse = TRUE` explicitly.

Intervals for every probability are formed on the logit scale by the delta method. Since $d \logit(p)/dp = 1/\{p(1-p)\}$,

$$se_{\logit}(\hat{p}) = \frac{se(\hat{p})}{\hat{p}(1-\hat{p})}, \quad CI = \logit\left(\logit(\hat{p}) \pm t_{0.975,\nu} se_{\logit}(\hat{p})\right),$$

the interval above keeps intervals inside $(0, 1)$ without truncation and improves coverage near the boundary. Estimates within 10^{-3} of a boundary have no meaningful logit-scale standard error and receive Wald intervals truncated to $[0, 1]$, flagged in the delivered tables.

```
design <- build_rep_design(dat, cfg)
des <- design$des
rep_des <- design$rep_des
nu <- degf(rep_des)
crit <- qt(0.975, df = nu)

glue("Design: {ncol(rep_des$repweights)} replicates, ",
     "{nu} degrees of freedom.")
```

Design: 84 replicates, 81 degrees of freedom.

```
# Parameter bookkeeping, in the order flatten_fit() produces.
param_meta <- bind_rows(
  tibble(kind = "pi", item = NA_character_, category = NA_integer_,
    segment = seq_len(K_star)),
  imap(cats, function(Cj, item_name)
    expand_grid(segment = seq_len(K_star), category = seq_len(Cj)) |>
    transmute(kind = "rho", item = item_name, category, segment)) |>
  list_rbind())

flatten_fit <- function(fit) c(fit$pi, unlist(map(fit$rho, as.vector)))

# maxit matches the full-sample fit. It was 5000 there and 1000 here for no
# stated reason, which is a difference in how hard the two halves of one
# estimate are tried: a replicate that stops at the cap enters the variance as
# a systematic deviation, which is the bias the matched tolerance above exists
# to prevent. The fit is checked rather than assumed.
theta_fun <- function(w_rep) {
  fit <- em_run(inp$Y, inp$OH, cats, w_rep, K_star,
    init = fit_w[c("pi", "rho")], maxit = 5000L, tol = 1e-10)
  stopifnot(isTRUE(fit$converged))
  flatten_fit(aligned_to(fit, fit_w))
}

est_vec <- flatten_fit(fit_w)
se_vec <- sqrt(diag(replicate_variance(rep_des, theta_fun, est_vec)))

ci_tbl <- param_meta |>
  mutate(estimate = as.numeric(est_vec), se = as.numeric(se_vec),
    boundary = estimate < 1e-3 | estimate > 1 - 1e-3,
    p_c = pmin(pmax(estimate, 1e-6), 1 - 1e-6),
    gse = se / (p_c * (1 - p_c)),
    lo = if_else(boundary, pmax(0, estimate - crit * se),
      plogis(qlogis(p_c) - crit * gse)),
    hi = if_else(boundary, pmin(1, estimate + crit * se),
      plogis(qlogis(p_c) + crit * gse))) |>
  select(-p_c, -gse)
```

10 Fit diagnostics

The chosen model is diagnosed before it is named or interpreted. Three questions are asked in order: does each item separate the segments, does each item earn the parameters it costs, and does the local independence assumption of the mixture above hold pairwise. A model that fails these is refit at a different K or with items removed, and only a model that passes proceeds to Section 12.

Two questions get asked of every item. The first is how much it separates the segments, measured as the mean over segment pairs of the total variation distance between their conditional response distributions:

$$D_j = \frac{2}{K(K-1)} \sum_{k < l} \frac{1}{2} \sum_{c=1}^{C_j} | \rho_{jck} - \rho_{jcl} | .$$

The inner expression is total variation distance between two discrete distributions, bounded in $[0, 1]$ and equal to the largest difference in probability the two assign to any event; the leading factor is the reciprocal of the number of pairs, $K(K-1)/2$, so D_j is a mean of quantities each in $[0, 1]$ and is itself in $[0, 1]$. A value of zero means every segment answers item j identically, making it a candidate for removal in a sensitivity refit; values near one mean the item alone nearly determines segment membership. Because it is a distance between distributions it is bounded in $[0, 1]$ whatever the number of response categories, so a binary item and a seven-point item are on the same scale. It is also sensitive to shape rather than only to location: a segment that avoids the middle of a scale registers here even when its average answer sits where everyone else's does, which suits a model that treats the categories as unordered.

The second question is how far the expected response actually travels, reported alongside as `range` and scaled by the number of categories so a two-point and a four-point item are comparable. This assumes the categories are ordered, which the model does not, but it is what an analyst reads off a profile plot. The two usually agree. Where they do not, the item is doing something worth looking at.

Table 4: Two views of how much each item separates the segments.

Item	Discrimination	Range
cash_transfer	0.467	0.884
computer	0.388	0.773
ran_out_food	0.340	0.632
govt_aid	0.284	0.565
home_internet	0.281	0.543
crime_victim	0.277	0.536
emigrate	0.270	0.505
own_finances	0.266	0.280
water_worry	0.245	0.450
feel_unsafe	0.162	0.189
refrigerator	0.126	0.230

Local independence is probed pairwise. For items a and b , let $\hat{P}_{ab}(c, d)$ be the design-weighted observed proportion giving response c to item a and d to item b , and let

$$\Pi_{ab} = \rho_a \text{diag}(\pi) \rho_b^\top$$

be the model-implied joint distribution, where ρ_a is the $C_a \times K$ matrix of conditional probabilities for item a . the model-implied table above is the model's own statement about the pair: condition on segment, multiply the two conditional distributions because the model asserts they are independent given segment, and average over segments weighted by prevalence. The bivariate residual is the total variation distance between observed and implied,

$$\text{BVR}_{ab} = \frac{1}{2} \sum_{c,d} | \hat{P}_{ab}(c, d) - \Pi_{ab}(c, d) | \in [0, 1].$$

This departs from the Pearson-type statistic conventionally reported, and deliberately. D. L. Oberski, G. H. van Kollenburg, and J. K. Vermunt [24] showed that the Pearson statistic's null distribution is already not chi-square for unweighted binary data, recommending bootstrap calibration; under design weighting and a pseudo-likelihood on a clustered sample no reference distribution is valid at all. A Rao-Scott design

adjustment would repair the design component but not the fact that the residual's null is not chi-square even under simple random sampling, so it does not rescue the statistic here. A statistic without a null distribution is an effect size, and an effect size should sit on an interpretable scale: the residual above is bounded in $[0, 1]$, is exactly zero under perfect local independence, reads as the largest probability discrepancy over any event in the two-way table, and shares its scale with the discrimination formula above.

The bound is not the same thing as comparability, and the distinction matters for how the table below is read. Every cell of the observed table carries sampling noise, so the residual a correctly specified model produces grows with the number of cells: a four-category item crossed with a binary one has eight cells and sits higher under the null than two binary items with four. The bound holds regardless; the null does not. The table therefore prints the cell count beside the residual, and a pair should be compared against pairs of the same table size rather than read off the ranking alone. A design-respecting reference — simulating from the fitted model under the realised cluster structure and reading each residual against its own simulated distribution — would remove the need for that caution and is the obvious extension; it is not done here. It also removes a numerical failure of the Pearson form, which divides by the model-implied cell probability and so can be dominated by a single near-empty corner of a large table.

```
bvr_tbl <- bvr_pairs(dat, w_vec, cfg$items, fit_w)

# Cells is printed beside the residual because the two are not independent.
# The bound [0, 1] holds whatever the table size, but the value the residual
# takes under a correctly specified model does not: every cell carries sampling
# noise, so a table with more cells accumulates more of it and sits higher under
# the null. Ranking an 8-cell pair against a 4-cell pair on the raw value ranks
# the table sizes as well as the dependence.
bvr_tbl |>
  mutate(cells = as.integer(cats[item_a]) * as.integer(cats[item_b]),
         bvr = round(bvr, 3)) |>
  slice_head(n = 10) |>
  select(item_a, item_b, cells, bvr) |>
  lca_table(align = "llrr", col.names = c("Item A", "Item B", "Cells", "BVR"),
            widths = c("13em", "13em", "5em", "5em"),
            caption = "Ten largest bivariate residuals, with the size of the
            two-way table each was computed on. A ranking, not a test, and not
            a ranking that is comparable across table sizes.")
```

Table 5: Ten largest bivariate residuals, with the size of the two-way table each was computed on. A ranking, not a test, and not a ranking that is comparable across table sizes.

Item A	Item B	Cells	BVR
feel_unsafe	crime_victim	8	0.057
ran_out_food	water_worry	4	0.051
own_finances	feel_unsafe	12	0.036
crime_victim	emigrate	4	0.031
refrigerator	home_internet	4	0.029
computer	home_internet	4	0.023
refrigerator	ran_out_food	4	0.023
ran_out_food	emigrate	4	0.021
water_worry	own_finances	6	0.021
computer	emigrate	4	0.021

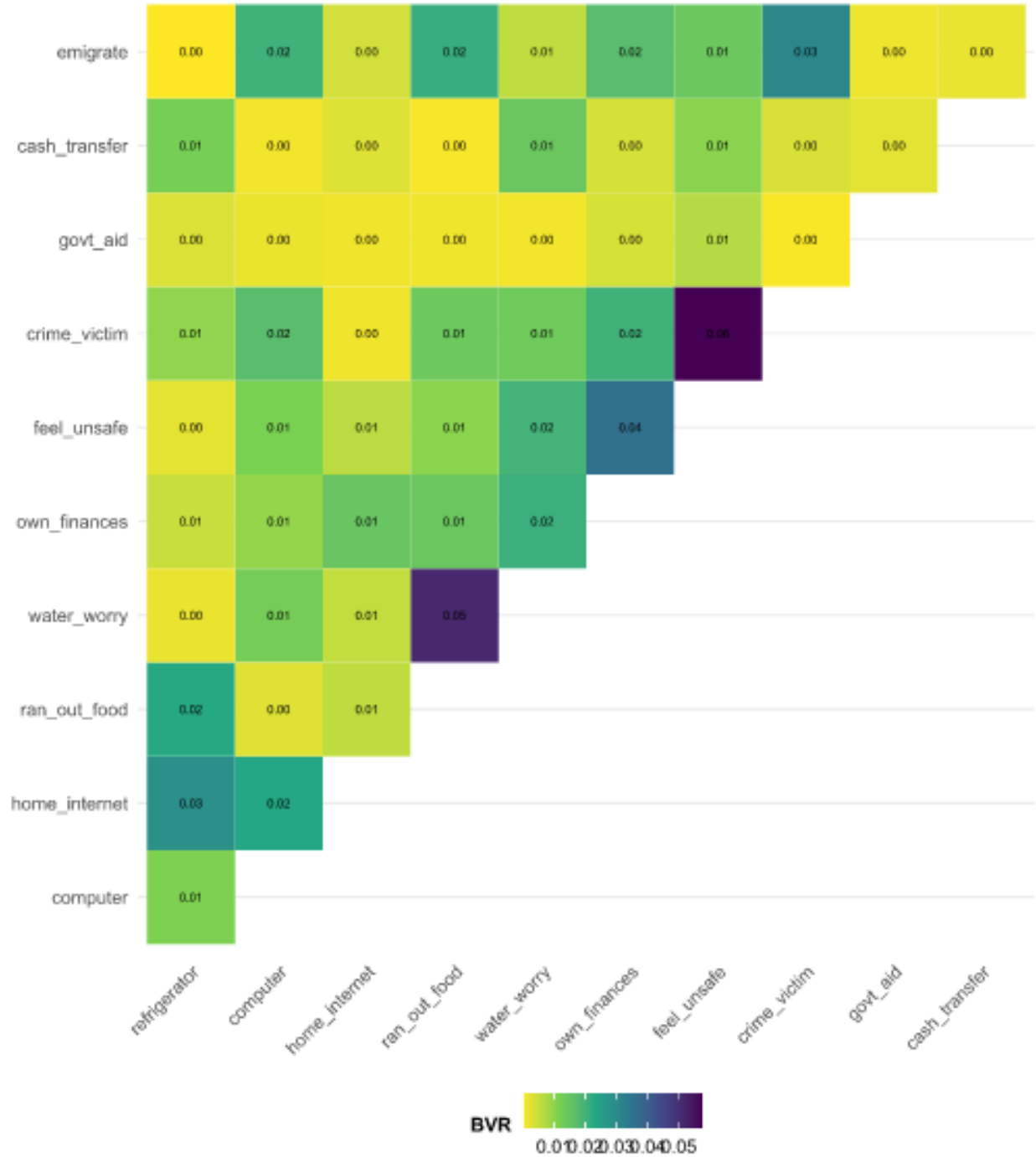


Figure 3: The same residuals as a heatmap. Darker cells are item pairs whose association the segments explain least well. Persistent dark pairs argue for dropping or merging an item, or trying $K+1$.

11 Validation against an independent implementation

At unit weights, the pseudo-log-likelihood above reduces to the ordinary log-likelihood, so the weighted EM and poLCA [25] maximize the same function and must agree up to optimizer tolerance and mode selection. This is enforced as a runtime check rather than asserted.

The comparison fits a third candidate by warm-starting our own EM at the polCA solution, so whichever basin attains the higher log-likelihood is retained and the comparison is between optima rather than between stopping rules. Where a gap appears, the two log-likelihoods separate the possible causes: equal log-likelihoods with different parameters indicate drift along a flat ridge under the stopping rule, a higher log-likelihood here indicates an inferior polCA mode, and a higher log-likelihood there indicates our own search needs more starts.

This check validates estimation. It says nothing about the variance implementation of Section 9, which is a separate concern and a reason to inspect the replicate diagnostics rather than treat agreement here as global.

```
polca_df <- dat |> select(all_of(cfg$items)) |> as.data.frame()
polca_f <- as.formula(paste0("cbind(", paste(cfg$items, collapse = ", "), ") ~ 1"))

set.seed(cfg$seed + 7L)
pl <- polCA(polca_f, data = polca_df, nclass = K_star, nrep = 10,
            maxiter = 5000, na.rm = TRUE, verbose = FALSE)
pl_aligned <- align_to(list(pi = as.numeric(pl$P), rho = map(pl$probs, t),
                           post = pl$posterior), fit_w)

fit_from_pl <- em_run(inp$Y, inp$OH, cats, rep(1, n_obs), K_star,
                    init = list(pi = pl_aligned$pi, rho = pl_aligned$rho),
                    maxit = 5000L, tol = 1e-10)
fit_unit <- if (fit_from_pl$ll > fit_u$ll) align_to(fit_from_pl, fit_w) else fit_u

validation <- tibble(
  max_pi_gap = max(abs(fit_unit$pi - pl_aligned$pi)),
  max_rho_gap = max(map2_dbl(fit_unit$rho, pl_aligned$rho, \(a, b) max(abs(a - b))))
validation$match <- max(validation$max_pi_gap, validation$max_rho_gap) < 0.005

cat(sprintf(paste0("VALIDATION (unit weights, our EM vs polCA):\n",
  " max |prevalence gap| = %.5f\n",
  " max |conditional-probability gap| = %.5f\n",
  " log-likelihood ours = %.4f | polCA = %.4f\n -> %s\n"),
  validation$max_pi_gap, validation$max_rho_gap, fit_unit$ll, pl$llik,
  if (validation$match) "MATCH: implementations agree."
  else "MISMATCH: investigate before trusting the weighted fit.))
```

```
VALIDATION (unit weights, our EM vs polCA):
max |prevalence gap| = 0.00000
max |conditional-probability gap| = 0.00000
log-likelihood ours = -9691.4554 | polCA = -9691.4554
-> MATCH: implementations agree.
```

12 Naming the segments

Segment naming is drafted by a language model under a fixed instrument and is a convenience with an audited override path, not a measurement claim. Recent evidence suggests language models perform annotation-adjacent tasks reliably when tightly instructed [26], [27], which is the narrow basis on which the

step is used here. In TSE terms it is a processing step (Section 2): it occurs after every number is computed, produces a display string, and never enters a computation, so a wrong name is a presentation error rather than a statistical one.

The instrument comprises a persona and rule set directing interpretation strictly from the displayed conditional response probabilities and item wording, one isolated call per segment, an optional survey-context paragraph fenced to referent resolution only, and a collision-gated harmonization pass that may edit labels but never descriptions. Isolation is empirical: joint prompts, in which the model sees all segments at once and must assign names one-to-one, confused near-neighbor segments and allowed a single confusion to corrupt two labels.

Two models carry the instrument, and the split follows from that isolation. Drafting is the **worker's**: one segment, one prompt, K times, with nothing to reason across and a fixed shape of answer, which is what a small model does reliably and cheaply. Harmonization is the **editor's**, because the pass is defined by seeing every label at once, which is the one thing the drafting calls exist to prevent, and because the collision gate means it fires rarely enough that a larger model costs little over a render. `cfg$llm_model_worker` and `cfg$llm_model_editor` name them, one id per endpoint; an empty editor entry runs both roles on the worker and the setup chunk prints that it has fallen back. The freeze file records which model drafted each label and, when the pass ran, which one edited it, so a name can be attributed to the model that produced it rather than to the workflow in general.

What decides whether the editor is called at all is a mechanical word-overlap test, and it is worth being clear about what that test can and cannot see. It normalises the labels, drops function words, strips suffixes crudely and compares word sets, which catches the collision that actually happens – one label being another with the words reordered – and does not catch a derivational pair such as *economic vulnerability* against *economically vulnerable*. It is a pre-filter, not a synonym detector. `cfg$label_harmonise` chooses the policy: "on_collision" runs the filter, "always" sends every drafted set to the editor once and does not depend on the filter being right. Because labels are frozen after the first render, "always" costs one extra call per output directory rather than one per render, which is the argument for it.

Reproducibility rests on a freeze file rather than a sampling seed. When `segment_labels.csv` exists in the output directory it is used and validated against K ; otherwise the model drafts once and writes it. Editing the file is how the analyst takes over naming, and deleting it triggers a redraft. A seed would make the draft reproducible only for a fixed model, endpoint, and package version; the file is reproducible unconditionally.

Labels are drafts. Naming happens only after the diagnostics of Section 10 and the validation of Section 11 have been read, and each label must be verified against its profile panel in Figure 6 before it is quoted.

===== SYSTEM PROMPT =====

You are a senior survey methodologist who reads latent class analysis (LCA) measurement models. In this work each latent class is called a SEGMENT; that is a word-choice preference and the statistical object is unchanged. Each segment is described only by its item-response probabilities: for every survey item, the probability that a member of that segment gives each answer. A segment leans toward the answers with high

probability. You interpret a segment strictly from these probabilities and the item wording, never from outside assumptions.

===== USER PROMPT (segment 1) =====

SURVEY CONTEXT

These items come from the 2023 AmericasBarometer survey of Ecuador, conducted by the LAPOP Lab at Vanderbilt University. Fieldwork was carried out face to face in Spanish by IPSOS between February and April 2023 with 1,604 respondents, drawn by a multi-stage probability design stratified by the three major regions of the country: Costa, Sierra, and Oriente.

The items analysed here record the material circumstances of the household and the respondent's own situation: whether the household owns particular durable goods, whether it has run short of food or water, whether finances and income have improved or worsened, whether the respondent feels unsafe or has been a victim of crime, whether the household receives government assistance or a conditional cash transfer, and whether the respondent intends to emigrate. The segments summarise patterns of economic vulnerability across these items.

ONE SEGMENT FROM A LATENT CLASS ANALYSIS (LCA) MEASUREMENT MODEL

SEGMENT 1 (estimated prevalence 21%):

refrigerator "Tiene un refrigerador (nevera) en el hogar"

P(No)=0.16, P(Sí)=0.84

computer "Tiene una computadora, computadora portátil, tableta, iPad en el hogar"

P(No)=0.74, P(Sí)=0.26

home_internet "Tiene acceso a internet en el hogar, incluyendo teléfono o tableta"

P(No)=0.27, P(Sí)=0.73

ran_out_food "Se ha quedado sin comida en los últimos 3 meses"

P(No)=0.20, P(Sí)=0.80

water_worry "Se ha quedado sin agua potable en los últimos 3 meses"

P(No)=0.27, P(Sí)=0.73

own_finances "Percepción de situación económica personal (En los últimos 12 meses)"

P(Mejor)=0.06, P(Igual)=0.09, P(Peor)=0.85

feel_unsafe "Percepción de seguridad en el vecindario"

P(Muy seguro(a))=0.03, P(Algo seguro(a))=0.15, P(Algo inseguro(a))=0.29, P(Muy inseguro(a))=0.53

crime_victim "Víctima del crimen en los últimos 12 meses"

P(Sí)=0.65, P(No)=0.35

govt_aid "Asistencia periódica en forma de dinero, alimentos o productos del gobierno"

P(Sí)=0.01, P(No)=0.99

cash_transfer "Beneficiario del programa de transferencias condicionadas"

P(Sí)=0.08, P(No)=0.92

emigrate "Intención de vivir/trabajar en otro país (próximos tres años)"

P(Sí)=0.62, P(No)=0.38

TASK

Read this single segment and return: a short DRAFT label (2 to 5 words) for an analyst to refine, and a one or two sentence factual description anchored to its high-probability answers. Each probability above belongs to one response category;

they are not yours to add together.

RULES:

1. Use only the numbers and item wording shown. Survey context only clarifies what the items refer to; attribute nothing that the numbers do not show.
 2. Quote a probability exactly as it is printed, for one response category at a time. Never add probabilities across categories and never describe a combined or total probability. If two adjacent answers both matter, name them separately with their own numbers.
 3. Anchor every statement to the items that stand out most.
 4. If nothing stands out, say the profile is diffuse rather than inventing a theme.
 5. Return only valid JSON: no prose before or after, no markdown fences.
- JSON (one object): {"label": "...", "description": "..."}

```
segment_labels <- get_segment_labels(fit_w, dictionary, cfg$items, cfg)

lca_table(segment_labels, widths = c("2em", "11em", "28em"),
  caption = "Segment labels and descriptions in use. Drafts, to verify against
the response profiles below.")
```

Table 6: Segment labels and descriptions in use. Drafts, to verify against the response profiles below.

K	Label	Description
1	High Vulnerability and Insecurity	This segment is characterized by severe material hardship, with high probabilities of having run out of food P(Si)=0.80 and water P(Si)=0.73, and a perception that personal finances have worsened P(Peor)=0.85. Members report high levels of insecurity, feeling very unsafe P(Muy inseguro(a))=0.53 and having been victims of crime P(Si)=0.65, while expressing a strong intention to emigrate P(Si)=0.62.
2	Secure Assets, High Instability	This segment is characterized by high ownership of a refrigerator P(Si)=0.99 and home internet P(Si)=1.00, while reporting a high intention to emigrate P(Si)=0.44 and a lack of government aid P(No)=0.98.
3	Economically Vulnerable and Insecure	This segment is characterized by a lack of computers P(Si)=0.01 and government aid P(Si)=0.01, combined with a perception that personal finances have worsened P(Peor)=0.64 and a high sense of neighborhood insecurity P(Muy inseguro(a))=0.31.
4	Subsidized Households	This segment is characterized by a very high probability of receiving conditional cash transfers P(Si)=0.95 and a high probability of receiving government aid P(Si)=0.58. Members frequently report their personal finances have worsened P(Peor)=0.56 and often feel either somewhat insecure P(Algo inseguro(a))=0.36 or very insecure P(Muy inseguro(a))=0.33 in their neighborhood.

```
lab_lev <- str_wrap(paste0("S", segment_labels$K, ": ", segment_labels$Label), 22)
lab_of <- function(k) factor(lab_lev[k], levels = lab_lev)
lab_line <- function(k) paste0("S", k, ": ", segment_labels$Label[k])
lab_short <- function(k) paste0("S", k)
```

13 The measurement model

Table 1

Table 7: Segment prevalences, design-weighted with 95 percent JK_n intervals, against the unweighted fit.

Segment	Weighted prevalence [95% CI]	Unweighted	Gap
S1: High Vulnerability and Insecurity	0.208 [0.145, 0.289]	0.209	-0.001
S2: Secure Assets, High Instability	0.500 [0.429, 0.570]	0.499	+0.001
S3: Economically Vulnerable and Insecure	0.192 [0.117, 0.300]	0.193	-0.000
S4: Subsidized Vulnerable Households	0.100 [0.059, 0.166]	0.100	+0.000

The figure below is the measurement model drawn out, and it is the latent class counterpart of a path diagram: one node for the latent variable, one label per item, and an edge whose weight stands where a loading would.

The number on each edge is that item's discrimination, and it reads as an average probability gap. A value of 0.85 means that two segments picked at random differ, on average, by 85 percentage points in how they answer that item, so the item nearly determines membership on its own. A value of 0.14 means they barely differ and the item is carrying almost nothing. The scale runs from 0 to 1 whatever the number of response categories, which is what makes a binary item and a four-category item comparable on the same picture. Edge thickness encodes the same number, so the diagram can be read at a glance and the labels checked afterwards.

What the edges cannot show is that these are estimates. The intervals for every conditional probability behind them are in the exported measurement model file.

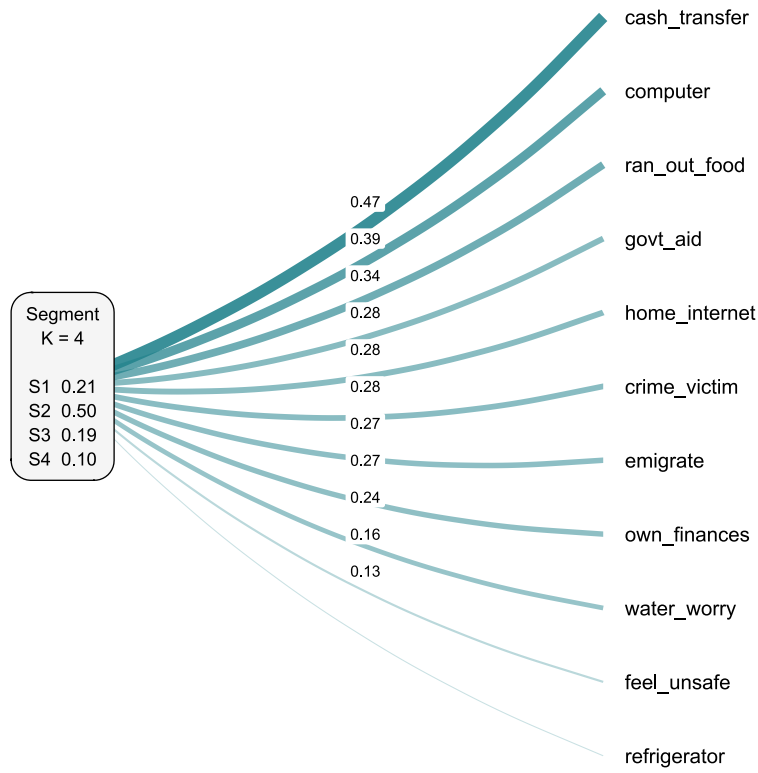


Figure 4: The measurement model. Each edge carries that item's discrimination: the average probability gap between two segments on it, from 0 to 1.

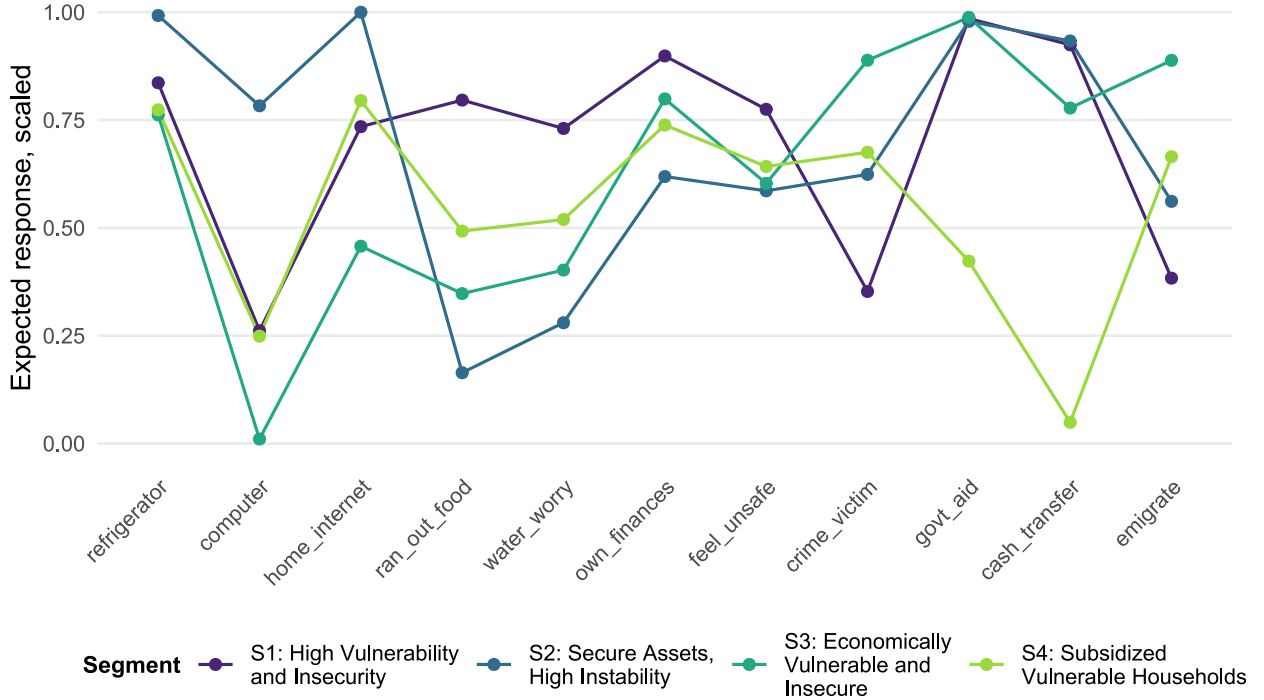


Figure 5: Segment profiles. Each item’s expected response is scaled to run from 0 to 1 so a binary and a four-category item are comparable. Lines that stay parallel mean the segments differ only in how high they answer overall, which a single continuum would describe. Lines that cross mean the segments reorder the items, which is what a class model recovers and a factor model cannot.

The same comparison as one number. Write m_{kj} for segment k ’s expected response to item j , rescaled to $[0, 1]$ so that a binary and a four-category item contribute the same possible range; without that rescaling the ratio is an artifact of the response formats. Decompose it,

$$m_{kj} = \underbrace{m^-}_{\text{level}} + \underbrace{(\ell_k - m^-)}_{\text{item}} + \underbrace{(\mu_j - m^-)}_{\text{item}} + \underbrace{e_{kj}}_{\text{pattern}}, \quad e_{kj} = m_{kj} - \ell_k - \mu_j + m^-,$$

where ℓ_k is segment k ’s mean over items, $\mu_j = \sum_k \pi_k m_{kj}$ is the item’s prevalence-weighted mean over segments, and $m^- = \sum_k \pi_k \ell_k$. The ratio reported below is the prevalence-weighted spread of the pattern term over the prevalence-weighted spread of the level term. Well under one says the segments differ mainly in how high they answer overall, which is a continuum, and a factor model would describe it in a fraction of the parameters. Near or above one says they reorder the items, which is structure no single factor can hold. It is evidence about which arm this battery calls for rather than a test, and it is read against the eigenvalue evidence in the factor report.

Two things about that construction are corrections to earlier versions of this document, and both move the number. The pattern term is the segment-by-item interaction e_{kj} , not the deviation $m_{kj} - \ell_k$ of an item from its own segment’s level. The deviation still carries the item main effect $\mu_j - m^-$ — the fact that some questions are endorsed more than others — which is a property of the battery and says nothing about whether the segments reorder anything. Left in, the statistic is strictly positive under the only null it claims to read on: when $m_{kj} = \ell_k + \mu_j$ exactly, every segment answers in the same order, $e_{kj} = 0$, and the ratio should be zero, but the deviation form returns $\text{sd}(\mu_j)/\text{sd}(\ell_k)$, which is unbounded. It also

moved when a constant was added to one item across every segment, which must not change the answer. Separately, both spreads and both marginals are now weighted by segment prevalence: unweighted, a segment holding four per cent of the population counted as much as one holding forty, so the ratio tracked the size of the smallest segment rather than the shape of the battery. On this fit the deviation form gives 3.78 unweighted and 5.9 weighted, against 4.3 for the interaction form reported here. The reading does not change; the number does, and only the last of the three is measuring what the sentence above says it measures.

The ratio is descriptive and comparable within a battery, not across batteries: the interaction residuals sum to zero over segments and over items, so their spread depends on both counts. Nothing selects on it and no threshold is attached to it.

Table 8: Level against pattern. Well under one is a continuum a factor model would describe more cheaply; near or above one is structure a single factor cannot hold.

SD of segment levels	SD of item deviations	Ratio
0.045	0.19	4.223

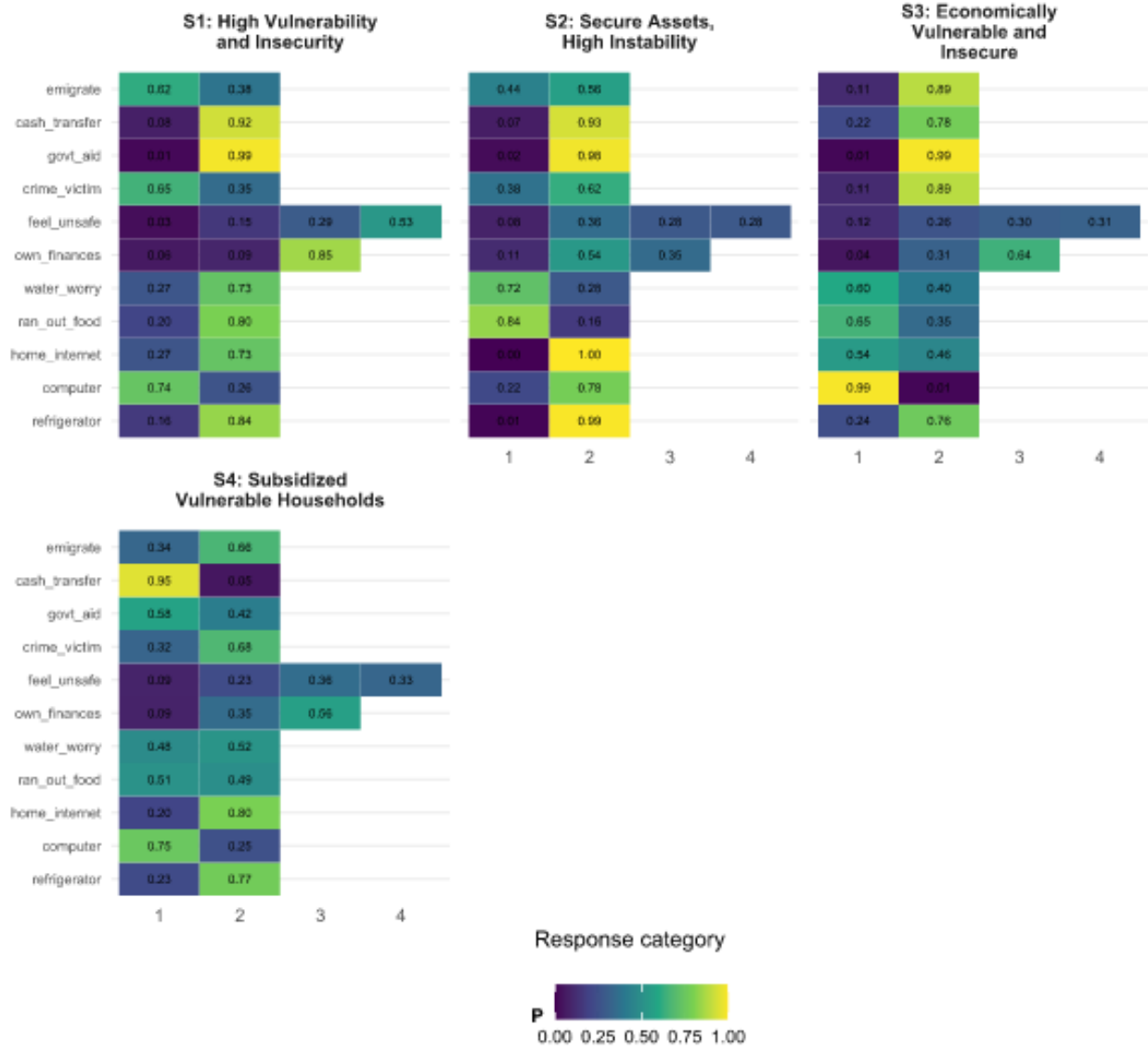


Figure 6: Conditional response probabilities $P(\text{response} \mid \text{segment})$ for every item, the parameters estimated by the formula above. Rows are items, columns are response categories.

14 Scoring segment membership

Local independence makes assignment exact rather than approximate for item-partial respondents. A missing item drops out of the within-segment product, so for respondent i with answered item set A_i ,

$$p_{ik}^{\text{obs}} = \frac{\pi_k \prod_{j \in A_i} \rho_{j y_{ij} k}}{\sum_l \pi_l \prod_{j \in A_i} \rho_{j y_{ij} l}}, \quad \hat{X}_i = \arg \max_k p_{ik}^{\text{obs}},$$

the posterior above is the exact posterior given what respondent i actually answered, under the model and under the condition $R_i \perp X_i \mid \mathbf{y}_{\text{obs},i}$: response propensity is conditionally independent of segment given the answers the respondent did give. That is stronger than it sounds and it is an assumption rather than a consequence of local independence. If the probability of skipping an item depends on segment directly,

the factor $P(R_i | X_i = k)$ does not cancel from the ratio above and the posterior is pulled toward π . The paragraph below argues that item nonresponse tracks education and age, which are themselves associated with segment, so the condition is unlikely to hold exactly here; the practical consequence is small only because almost every partial responder is missing a single item. An information floor `cfg$min_items` is enforced: respondents answering fewer items receive no assignment rather than a weakly informed one.

Scoring precedes domain estimation, and all domain estimation runs on the scored frame rather than the estimation sample. This follows the logic of three-step latent class analysis, in which the assignment step covers every respondent it validly can, and it matters substantively rather than cosmetically: item nonresponse on an attitude battery correlates with education, age, and other characteristics of interest, so restricting domain estimation to complete responders introduces a selection into exactly the composition being estimated. This is the item-level nonresponse concern of Section 2.

The cost, and it is the correlated-error case noted there, is that respondents scored from fewer items carry flatter posteriors and therefore more classification error than those scored from all items. The replicate design is rebuilt on the scored frame and the singleton stratum check is repeated there.

```
pred <- predict_segments(survey_dat_full, fit_w, cfg$items, cfg$min_items)
survey_scored <- bind_cols(survey_dat_full, pred)
post_cols <- paste0("post_segment", seq_len(K_star))

# A demographic level whose last respondent went unscored has to go with them:
# svyby returns no row for an empty level while count(.drop = FALSE) below
# returns one, and the three estimator frames would stop lining up.
scored <- survey_scored |>
  filter(!is.na(segment)) |>
  mutate(segment = factor(segment, levels = seq_len(K_star))) |>
  drop_empty_levels(cfg$aux, "scored frame")

design_s <- build_rep_design(scored, cfg)
des_s <- design_s$des
rep_des_s <- design_s$rep_des
nu_s <- degf(rep_des_s)
crit_s <- qt(0.975, df = nu_s)

glue(
  "Estimation sample: {n_obs} item-complete cases.\n",
  "Scored sample: {nrow(scored)} of {nrow(survey_scored)} respondents ",
  "(floor: {cfg$min_items} items), a gain of {nrow(scored) - n_obs}.\n",
  "Domain design: {ncol(rep_des_s$repweights)} replicates, ",
  "{nu_s} degrees of freedom.")
```

```
Estimation sample: 1445 item-complete cases.
Scored sample: 1604 of 1604 respondents (floor: 6 items), a gain of 159.
Domain design: 84 replicates, 81 degrees of freedom.
```

Table 9: Assignment quality by number of items answered. Respondents scored from fewer items carry flatter posteriors and so more classification error. If the mean falls sharply at the low end, raise `cfg$min_items`.

Items answered	n	Mean max posterior
6	1	0.607
7	8	0.817
8	10	0.772
9	37	0.801
10	103	0.798
11	1445	0.836

The same quantity by domain level is the diagnostic the BCH correction of Section 15 rests on, because one pooled classification table is inverted for every level. If assignment sharpness differs across the levels of a demographic, the classification error rates differ with it, and a single $\hat{D}$ is the wrong object for all of them. This is read before the corrected column is quoted; nothing is corrected on the strength of it and no threshold is attached.

Table 10: Assignment quality by domain level. The BCH correction inverts one classification table for every level of a demographic, which assumes the error rates are the same at each. Levels whose mean sits well below the others are where that assumption is weakest.

Variable	Level	n	Mean max posterior
age_cat	16-29	653	0.845
age_cat	30-44	420	0.832
age_cat	45-59	310	0.808
age_cat	60+	221	0.825
sex	Female	799	0.822
sex	Male	786	0.842
education	Primary	312	0.794
education	Secondary	798	0.816
education	Tertiary	447	0.890
urban	Rural	531	0.815
urban	Urban	1073	0.840
employment	Employed	697	0.843
employment	Homemaker	256	0.796
employment	Not in labor force	76	0.816
employment	Retired	88	0.849
employment	Student	182	0.885
employment	Unemployed	272	0.804

15 Domain estimation

The quantity of interest is segment prevalence within demographic domains. Writing d for a level of a demographic variable,

$$\hat{p}_{dk} = \frac{\sum_{i \in d} w_i \mathbf{1}(\hat{X}_i = k)}{\sum_{i \in d} w_i},$$

the design-weighted proportion of domain d assigned to segment k . the domain estimator above takes its variance from the same JK_n replicates as the jackknife variance above. In implementation this is `svyby` over the demographic with `svymean` of the segment factor inside each level, so the estimate and its variance come from the replicate design directly. Rows sum to one across segments by construction, which means the cells within a row are not independent: if one segment's share rises, others must fall.

15.1.a Three estimators

Each domain table carries the same quantity three ways, so the two corrections can be read separately. **Unweighted** is a plain cross-tab: sample proportions with binomial standard errors, as if the respondents had been drawn independently. **Design-based** applies the sampling weights and takes its variance from the JKn replicates. **BCH** additionally removes the attenuation that hard modal assignment introduces. The gap from the first row to the second is what ignoring the design costs; the gap from the second to the third is what modal assignment costs.

15.1.b The correction

This estimator uses hard modal assignment and is therefore **attenuated**. Classification error pulls composition toward the marginal distribution, so differences between demographic levels are understated [28], [29]. The BCH correction removes it [28]–[31]. Writing W_i for the modal assignment, the design-weighted classification table is

$$\hat{D}_{ks} = \frac{\sum_i w_i p_{ik} \mathbf{1}(W_i = s)}{\sum_i w_i p_{ik}},$$

the probability that a respondent truly in segment k is assigned to segment s . Each respondent's hard assignment is replaced by row W_i of $\hat{D}^{-1}$, so respondent i contributes the vector $w_i (\hat{D}^{-1})_{W_i}$, where the design-based estimator contributed a single indicator; the denominator of the domain estimator above is unchanged. Because $\hat{D}$ is row-stochastic, $\hat{D}\mathbf{1} = \mathbf{1}$ and hence $\hat{D}^{-1}\mathbf{1} = \mathbf{1}$, so the rows of the inverse sum to one and the corrected shares still sum to one across segments. Individual entries of $\hat{D}^{-1}$ are routinely negative, so a corrected share can fall outside $[0, 1]$. That is a property of the estimator rather than a fault, and such cells are reported untruncated and flagged rather than clipped to a boundary the estimator did not produce.

Two conditions on the correction have to be stated rather than assumed. One pooled $\hat{D}$ is inverted for every domain, which requires the classification error rates to be the same at every level of the demographic — non-differential classification with respect to the domain variable. Section 14 shows that respondents who answered fewer items carry flatter posteriors, and item nonresponse is not independent of age or education, so the condition is not guaranteed here; the by-level mean maximum posterior reported below is the diagnostic, and it has to be read before the corrected column is quoted. And $\hat{D}$ has to be well enough conditioned for its inverse to mean anything. An exactly singular table stops the run because `solve()` refuses it; a near-singular one — a segment that took very few modal assignments — does not, and the correction it produces is unstable rather than visibly wrong. The engine attaches the reciprocal condition number of $\hat{D}$ to every call, and replicates falling below the floor are counted and disclosed rather than dropped, which is the only honest treatment available: there is nothing to repair.

```
marginals <- map(cfg$aux, function(v) {
  m <- svymean(reformulate(v), rep_des_s, na.rm = TRUE)
  raw <- scored |>
    filter(!is.na(.data[[v]])) |>
    count(.data[[v]], name = "n") |>
    set_names(c("level", "n"))
  tibble(variable = v,
    level = str_remove(names(coef(m)), paste0("^", v)),
    weighted = as.numeric(coef(m)),
```

```

      se = as.numeric(SE(m))) |>
      left_join(mutate(raw, level = as.character(level)), by = "level")
    }) |>
    list_rbind()

# svyby splits the design by the demographic and runs svymean(~segment) within
# each level. Columns are read by position (by-variable, then K estimates, then
# K SEs) because svyby's naming for a factor outcome varies by version.
domain <- map(cfg$aux, function(v) {
  sb <- as.data.frame(svyby(~ segment, reformulate(v), rep_des_s, svymean,
                           na.rm = TRUE))

  vals <- sb[, -1, drop = FALSE]
  stopifnot(ncol(vals) == 2 * K_star)
  tibble(variable = v,
          level = rep(as.character(sb[[1]]), times = K_star),
          segment = rep(seq_len(K_star), each = nrow(sb)),
          p = as.numeric(as.matrix(vals[, seq_len(K_star)])),
          se = as.numeric(as.matrix(vals[, K_star + seq_len(K_star)])))
}) |>
  list_rbind() |>
  filter(!is.na(level)) |>
  mutate(lo = pmax(0, p - crit_s * se), hi = pmin(1, p + crit_s * se))

# BCH. Posteriors and modal assignment stay at their full-sample values; D and
# the correction weights are rebuilt inside every replicate, so the intervals
# carry the design variance of the classification table and the cross-tab
# together rather than treating D as known.
P_fixed <- as.matrix(scored[, post_cols, drop = FALSE])
modal_fixed <- as.integer(scored$segment)

dom_meta <- map(cfg$aux, function(v)
  expand_grid(level = levels(scored[[v]]), segment = seq_len(K_star)) |>
  mutate(variable = v) |>
  list_rbind() |>
  mutate(idx = row_number())

domain_theta <- function(w_rep) {
  wU = w_rep * bch_weights(P_fixed, modal_fixed, w_rep)
  unlist(map(cfg$aux, function(v) {
    M = outer(as.character(scored[[v]]), levels(scored[[v]]), "==" ) + 0
    M[is.na(M)] = 0
    as.vector(t(crossprod(M, wU) / as.numeric(crossprod(M, w_rep))))
  })), use.names = FALSE)
}

dom_est <- domain_theta(weights(des_s, "sampling"))
dom_se <- sqrt(diag(replicate_variance(rep_des_s, domain_theta, dom_est)))
stopifnot(length(dom_est) == nrow(dom_meta))

domain_bch <- dom_meta |>
  mutate(p = as.numeric(dom_est), se = as.numeric(dom_se),

```

```

      lo = p - crit_s * se, hi = p + crit_s * se) |>
select(variable, level, segment, p, se, lo, hi)

# Design-based estimator as a theta function, so the full replicate covariance
# between domain estimates exists. The separations are Wald contrasts on this
# covariance rather than an interval-overlap rule, which tested at an alpha
# well below its nominal level.
# The tests run on the design-based estimator because that estimator's
# intervals are the ones printed beside them, so a reader can check the test
# against a table in front of them. This is NOT a claim that the uncorrected
# test is the conservative one. An earlier version of this comment said modal
# assignment attenuates differences, so a pair resolved here "can only widen"
# under the correction. That does not follow: BCH moves the point difference
# and its standard error at once -- the inverse classification table has
# diagonal entries above one and off-diagonal entries that are routinely
# negative -- and the ratio of the two is not signed in general. Which test is
# the conservative one is an empirical property of this D, not of the method.
S_fixed <- outer(modal_fixed, seq_len(K_star), `==`) + 0

domain_theta_dsg <- function(w_rep) {
  unlist(map(cfg$aux, function(v) {
    M = outer(as.character(scored[[v]]), levels(scored[[v]]), "==" ) + 0
    M[is.na(M)] = 0
    as.vector(t(crossprod(M, w_rep * S_fixed) / as.numeric(crossprod(M, w_rep))))
  }), use.names = FALSE)
}

dom_est_dsg <- domain_theta_dsg(weights(des_s, "sampling"))
V_dsg <- replicate_variance(rep_des_s, domain_theta_dsg, dom_est_dsg)

# The same quantity svyby computed, by a different route; agreement validates
# the read-by-position handling of svyby's output.
chk <- dom_meta |>
  mutate(p_theta = dom_est_dsg) |>
  inner_join(select(domain, variable, level, segment, p),
    by = c("variable", "level", "segment"))
stopifnot(max(abs(chk$p_theta - chk$p)) < 1e-6)

wald_dsg <- list(meta = dom_meta, est = dom_est_dsg, V = V_dsg, df = nu_s)

# Naive estimates: unweighted proportions with binomial standard errors, as if
# the respondents were an iid sample. This is what a plain cross-tab gives, and
# it is here so the two corrections can be read separately: the gap to the
# design-based row is what ignoring the design costs, and the gap from there to
# the BCH row is what modal assignment costs.
domain_naive <- map(cfg$aux, function(v) {
  scored |>
    filter(!is.na(.data[[v]])) |>
    count(.data[[v]], segment, name = "n", .drop = FALSE) |>

```

```

set_names(c("level", "segment", "n")) |>
group_by(level) |>
mutate(n_level = sum(n), p = n / n_level) |>
ungroup() |>
transmute(variable = v, level = as.character(level),
          segment = as.integer(segment), p,
          se = sqrt(p * (1 - p) / n_level),
          # Logit-scale, like every other interval here. The three
          # estimators are compared on the width of their intervals,
          # in the text below and in the prompt of @sec-domain-read, so the
          # construction has to be held constant or the ratio is part
          # estimator and part transform. The 1.96 stays: this row is a
          # binomial standard error on a plain cross-tab and carries no
          # design degrees of freedom.
          p_c = pmin(pmax(p, 1e-3), 1 - 1e-3),
          gse = se / (p_c * (1 - p_c)),
          lo = plogis(qlogis(p_c) - qnorm(0.975) * gse),
          hi = plogis(qlogis(p_c) + qnorm(0.975) * gse)) |>
  select(-p_c, -gse)
}) |>
list_rbind() |>
arrange(variable, segment, level)

# The shift summaries difference these against each other. Ordering them the
# same way is not enough on its own: svyby returns nothing for a demographic
# level with no cases while count(.drop = FALSE) returns a row for it, so a
# positional difference can pair the wrong cells rather than fail. The frames are
# joined by key below and the row counts are checked here, which turns that into
# a halt.
domain <- arrange(domain, variable, segment, level)
domain_bch <- arrange(domain_bch, variable, segment, level)
stopifnot(nrow(domain) == nrow(domain_naive), nrow(domain) == nrow(domain_bch))

est_levels <- c("Unweighted", "Design-based", "BCH")

domain_both <- bind_rows(mutate(domain_naive, estimator = "Unweighted"),
                        mutate(domain, estimator = "Design-based"),
                        mutate(domain_bch, estimator = "BCH")) |>
  mutate(estimator = factor(estimator, levels = est_levels))

# Cells at an observed 0 or 1 have a zero binomial standard error, so the
# ratio there is 0/0. Those cells say nothing about the clustering cost and
# are excluded from the median rather than allowed to poison it.
se_ratio <- domain |>
  select(variable, level, segment, se) |>
  inner_join(select(domain_naive, variable, level, segment, se_naive = se),
            by = c("variable", "level", "segment")) |>
  transmute(ratio = se / se_naive) |>
  pull(ratio)

cat(sprintf(paste0("Weighting moves the point estimates by a median of %.4f;

```

```

the BCH\n",
"correction moves them a further %.4f, at most %.4f,
against a median\n",
"design-based standard error of %.4f. Read each shift
against that\n",
"standard error: one well inside it is not moving
anything the design\n",
"can resolve. The design-based standard errors are a
median of %.2f\n",
"times the naive ones, which is what ignoring the
clustering costs.\n"),
median(abs(domain$p - domain_naive$p)),
median(abs(domain_bch$p - domain$p)),
max(abs(domain_bch$p - domain$p)),
median(domain$se), median(se_ratio[is.finite(se_ratio)]))

```

```

Weighting moves the point estimates by a median of 0.0013;
the BCH
correction moves them a further 0.0173, at most 0.0877,
against a median
design-based standard error of 0.0214. Read each shift
against that
standard error: one well inside it is not moving
anything the design
can resolve. The design-based standard errors are a
median of 1.09
times the naive ones, which is what ignoring the
clustering costs.

```

Before anything is sent, R tests every pair of levels within each segment: a design-based Wald test on the difference, using the replicate covariance between the two estimates, Holm-adjusted across the pairs within the demographic. The model is handed the list of pairs that resolve and told to name only those. The tests run on the design-based estimator because its intervals are the ones printed in the tables above, so a reader can check a listed pair against numbers in front of them. That is a reporting choice and not a claim that the uncorrected test is the conservative one: the correction moves the difference and its standard error together, the inverse classification table carries diagonal entries above one and negative entries off it, and the ratio of the two is not signed in general. Whether correcting strengthens or weakens a given contrast is an empirical property of this $\hat{D}$, and neither direction is asserted here.

```

# The BCH intervals above hold the step-one measurement model fixed. This
# refits the model inside every replicate and lets the replicate's own
# posteriors drive the classification table and the cross-tab, so step-one
# uncertainty enters the variance. One demographic keeps the cost at one EM
# refit per replicate; the ratio, not the coverage, is the deliverable.
idx_est <- which(scored$in_analysis)
Y_sc <- map(cfg$items, function(it) as.integer(scored[[it]]))
aux_sl <- cfg$aux[1]

```

```

theta_step1 <- function(w_rep) {
  fit_r = align_to(em_run(inp$Y, inp$OH, cats, w_rep[idx_est], K_star,
    init = fit_w[c("pi", "rho")],
    maxit = 5000L, tol = 1e-10), fit_w)
  post_r = posterior_of(fit_r$pi, fit_r$rho, Y_sc)
  modal_r = max.col(post_r, ties.method = "first")
  wU = w_rep * bch_weights(post_r, modal_r, w_rep)
  unlist(map(aux_s1, function(v) {
    M = outer(as.character(scored[[v]]), levels(scored[[v]]), "==") + 0
    M[is.na(M)] = 0
    as.vector(t(crossprod(M, wU) / as.numeric(crossprod(M, w_rep))))
  }), use.names = FALSE)
}

est_s1 <- theta_step1(weights(des_s, "sampling"))
Wm_s <- weights(rep_des_s, type = "analysis")
Theta_s1 <- do.call(rbind, future_map(seq_len(ncol(Wm_s)), function(r) {
  out = try(theta_step1(Wm_s[, r]), silent = TRUE)
  if (inherits(out, "try-error")) rep(NA_real_, length(est_s1)) else out
}), .options = furrr_options(seed = NULL)))
# A failed replicate is counted and halts, never quietly dropped. Dropping a
# deviation can only narrow the spread, so a silently shortened sum understates
# the very quantity this chunk exists to measure, and it does so in the
# direction that makes the answer look better.
n_failed <- sum(!complete.cases(Theta_s1))
if (n_failed > 0)
  stop(n_failed, " of ", nrow(Theta_s1), " replicates failed to refit. The ",
    "step-one sensitivity cannot be reported from the survivors, because ",
    "dropping them narrows the spread it is measuring.")
d_s1 <- sweep(Theta_s1, 2, est_s1, "-")
se_s1 <- sqrt(diag(rep_des_s$scale * crossprod(d_s1 * sqrt(rep_des_s$rscales))))

s1_tbl <- dom_meta |>
  filter(variable %in% aux_s1) |>
  mutate(se_step1 = se_s1) |>
  inner_join(select(domain_bch, variable, level, segment, p, se),
    by = c("variable", "level", "segment")) |>
  mutate(ratio = se_step1 / se)

glue("Replicates converged: {nrow(Theta_s1) - n_failed} of {nrow(Theta_s1)}. ",
  "Propagating step one multiplies the BCH standard errors for {aux_s1} ",
  "by a median of {round(median(s1_tbl$ratio, na.rm = TRUE), 2)} ",
  "and at most {round(max(s1_tbl$ratio, na.rm = TRUE), 2)}.")

```

Replicates converged: 84 of 84. Propagating step one multiplies the BCH standard errors for age_cat by a median of 2.17 and at most 4.22.

```

s1_tbl |>
  transmute(Segment = lab_line(segment), Level = level,

```

```

`SE fixed` = sprintf("%.4f", se),
`SE propagated` = sprintf("%.4f", se_step1),
Ratio = sprintf("%.2f", ratio)) |>
lca_table(align = "llrrr", widths = fit_widths(5, first = 2.4), font_size = 8,
caption = "BCH standard errors with the step-one measurement
model fixed against refit within every replicate, for
one demographic.")

```

Table 11: BCH standard errors with the step-one measurement model fixed against refit within every replicate, for one demographic.

Segment	Level	SE fixed	SE propagated	Ratio
S1: High Vulnerability and Insecurity	16-29	0.0198	0.0556	2.81
S2: Secure Assets, High Instability	16-29	0.0262	0.0606	2.31
S3: Economically Vulnerable and Insecure	16-29	0.0178	0.0310	1.74
S4: Subsidized Vulnerable Households	16-29	0.0161	0.0366	2.27
S1: High Vulnerability and Insecurity	30-44	0.0259	0.0507	1.96
S2: Secure Assets, High Instability	30-44	0.0331	0.0445	1.35
S3: Economically Vulnerable and Insecure	30-44	0.0283	0.0550	1.94
S4: Subsidized Vulnerable Households	30-44	0.0195	0.0405	2.07
S1: High Vulnerability and Insecurity	45-59	0.0328	0.0866	2.64
S2: Secure Assets, High Instability	45-59	0.0349	0.0591	1.69
S3: Economically Vulnerable and Insecure	45-59	0.0315	0.1329	4.22
S4: Subsidized Vulnerable Households	45-59	0.0225	0.0871	3.87
S1: High Vulnerability and Insecurity	60+	0.0334	0.1025	3.07
S2: Secure Assets, High Instability	60+	0.0419	0.0497	1.19
S3: Economically Vulnerable and Insecure	60+	0.0427	0.1168	2.73
S4: Subsidized Vulnerable Households	60+	0.0237	0.0333	1.41

15.2 age_cat

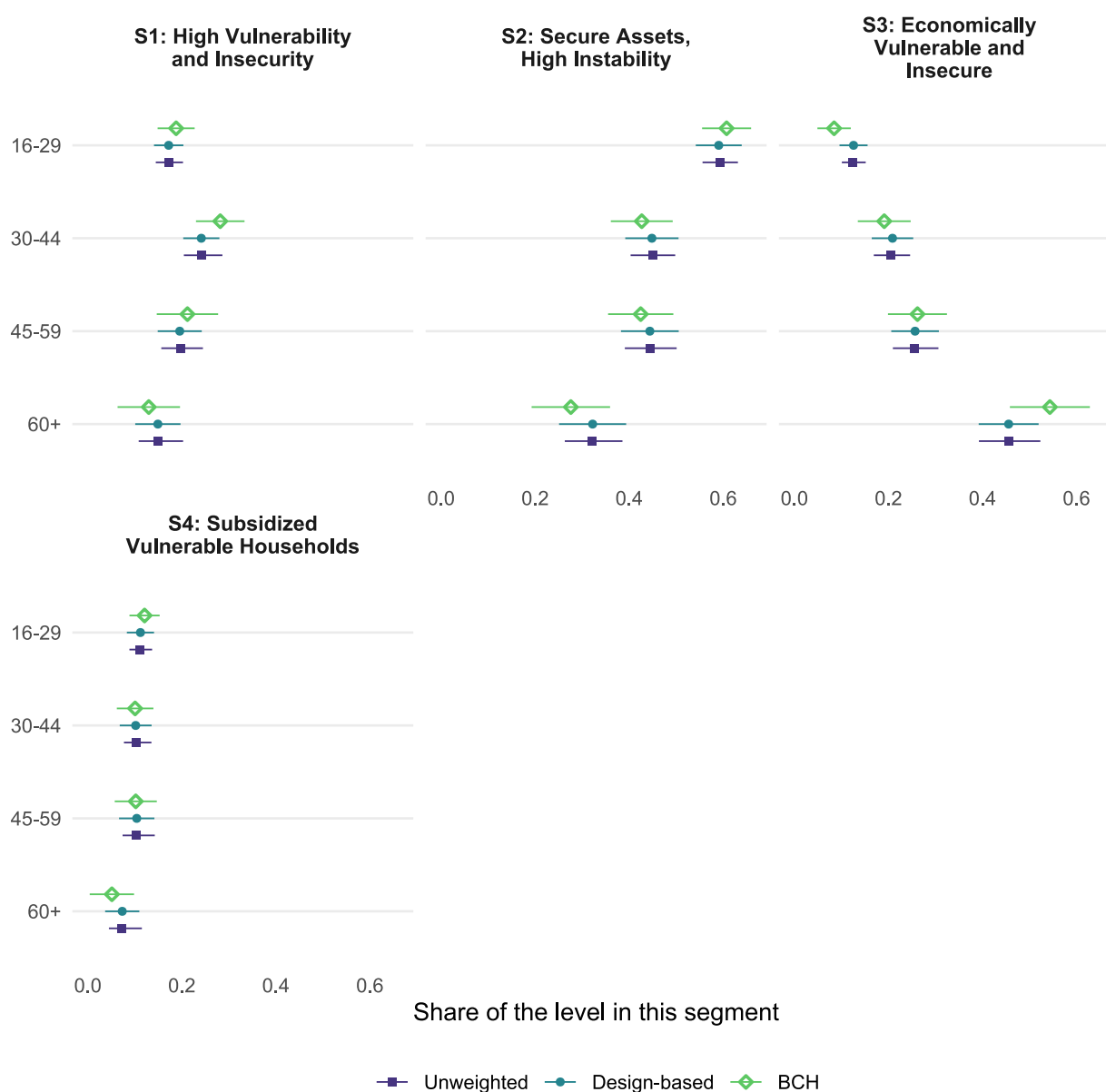
Table 12: Composition by age_cat: design-weighted proportion among respondents who answered the item, with its JK_n standard error.

Level	n	Share
16-29	653	0.405 (0.007)
30-44	420	0.261 (0.007)
45-59	310	0.195 (0.008)
60+	221	0.138 (0.007)

Table 13: Share of each level of age_cat falling in each segment under three estimators, with standard errors. Unweighted is a plain cross-tab with a binomial standard error. Design-based adds the weights and takes its variance from the JK_n replicates. BCH additionally removes the attenuation that modal assignment introduces. Columns sum to one down the segments. * fewer than 30 respondents; not a finding. † outside [0, 1], which the BCH correction permits and truncation would misstate.

Segment	Estimator	16-29	30-44	45-59	60+
S1: High Vulnerability and Insecurity	Unweighted	0.172 (0.015)	0.243 (0.021)	0.197 (0.023)	0.149 (0.024)
S1: High Vulnerability and Insecurity	Design-based	0.172 (0.016)	0.241 (0.019)	0.195 (0.024)	0.149 (0.024)
S1: High Vulnerability and Insecurity	BCH	0.188 (0.020)	0.282 (0.026)	0.212 (0.033)	0.130 (0.033)

Segment	Estimator	16-29	30-44	45-59	60+
S2: Secure Assets, High Instability	Unweighted	0.594 (0.019)	0.450 (0.024)	0.445 (0.028)	0.321 (0.031)
S2: Secure Assets, High Instability	Design-based	0.590 (0.025)	0.448 (0.028)	0.444 (0.031)	0.322 (0.036)
S2: Secure Assets, High Instability	BCH	0.607 (0.026)	0.427 (0.033)	0.425 (0.035)	0.276 (0.042)
S3: Economically Vulnerable and Insecure	Unweighted	0.124 (0.013)	0.205 (0.020)	0.255 (0.025)	0.457 (0.034)
S3: Economically Vulnerable and Insecure	Design-based	0.126 (0.015)	0.209 (0.022)	0.257 (0.025)	0.456 (0.032)
S3: Economically Vulnerable and Insecure	BCH	0.084 (0.018)	0.191 (0.028)	0.262 (0.032)	0.543 (0.043)
S4: Subsidized Vulnerable Households	Unweighted	0.110 (0.012)	0.102 (0.015)	0.103 (0.017)	0.072 (0.017)
S4: Subsidized Vulnerable Households	Design-based	0.112 (0.015)	0.102 (0.017)	0.104 (0.019)	0.073 (0.018)
S4: Subsidized Vulnerable Households	BCH	0.121 (0.016)	0.101 (0.020)	0.102 (0.022)	0.051 (0.024)



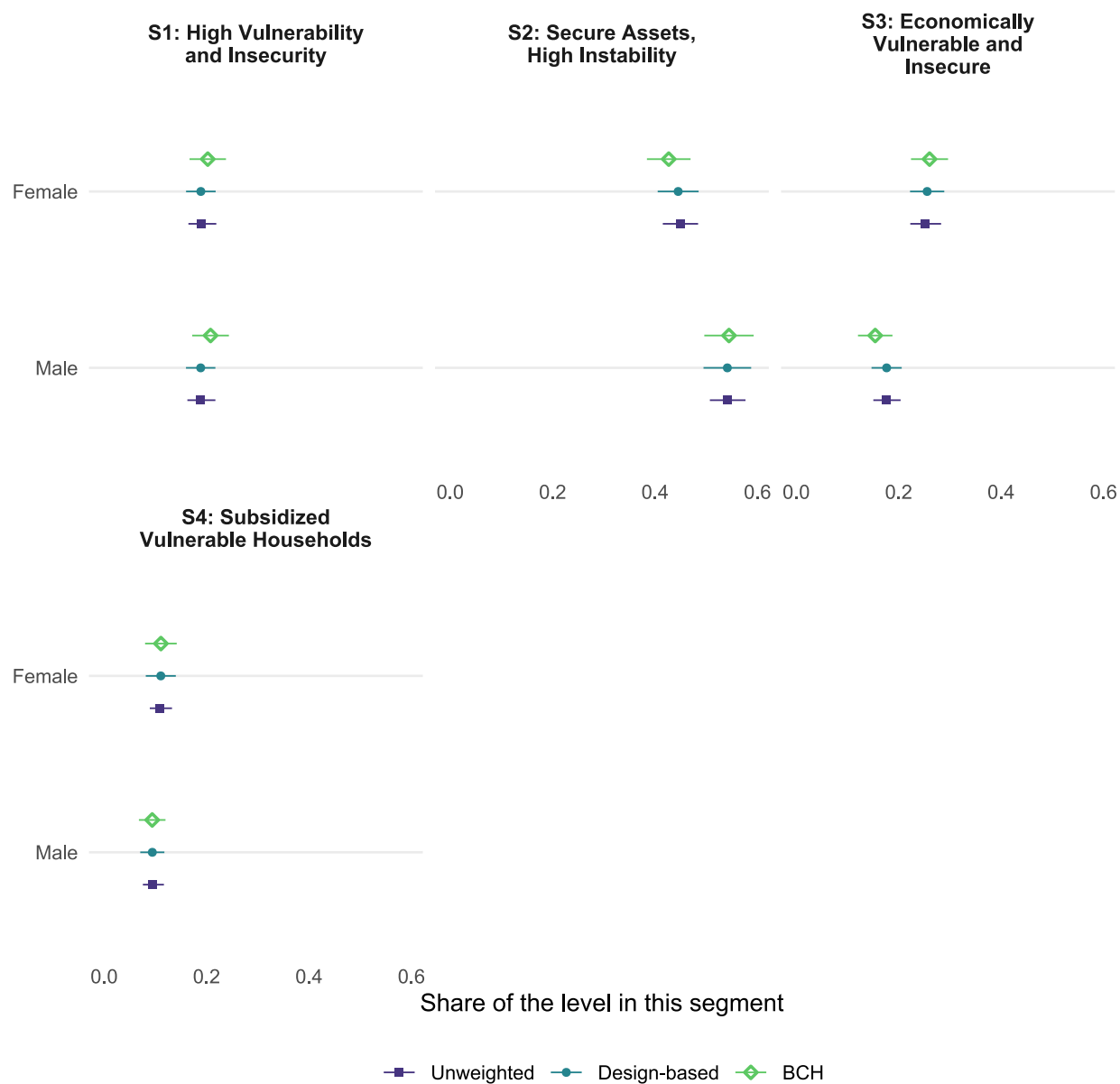
15.3 sex

Table 14: Composition by sex: design-weighted proportion among respondents who answered the item, with its JK_n standard error.

Level	n	Share
Male	786	0.499 (0.005)
Female	799	0.501 (0.005)

Table 15: Share of each level of sex falling in each segment under three estimators, with standard errors. Unweighted is a plain cross-tab with a binomial standard error. Design-based adds the weights and takes its variance from the JK_n replicates. BCH additionally removes the attenuation that modal assignment introduces. Columns sum to one down the segments. * fewer than 30 respondents; not a finding. † outside [0, 1], which the BCH correction permits and truncation would misstate.

Segment	Estimator	Female	Male
S1: High Vulnerability and Insecurity	Unweighted	0.190 (0.014)	0.188 (0.014)
S1: High Vulnerability and Insecurity	Design-based	0.189 (0.015)	0.188 (0.014)
S1: High Vulnerability and Insecurity	BCH	0.202 (0.018)	0.208 (0.018)
S2: Secure Assets, High Instability	Unweighted	0.449 (0.018)	0.542 (0.018)
S2: Secure Assets, High Instability	Design-based	0.445 (0.020)	0.541 (0.023)
S2: Secure Assets, High Instability	BCH	0.427 (0.021)	0.544 (0.024)
S3: Economically Vulnerable and Insecure	Unweighted	0.252 (0.015)	0.176 (0.014)
S3: Economically Vulnerable and Insecure	Design-based	0.256 (0.017)	0.176 (0.015)
S3: Economically Vulnerable and Insecure	BCH	0.260 (0.018)	0.154 (0.017)
S4: Subsidized Vulnerable Households	Unweighted	0.109 (0.011)	0.094 (0.010)
S4: Subsidized Vulnerable Households	Design-based	0.111 (0.015)	0.094 (0.012)
S4: Subsidized Vulnerable Households	BCH	0.111 (0.016)	0.094 (0.013)



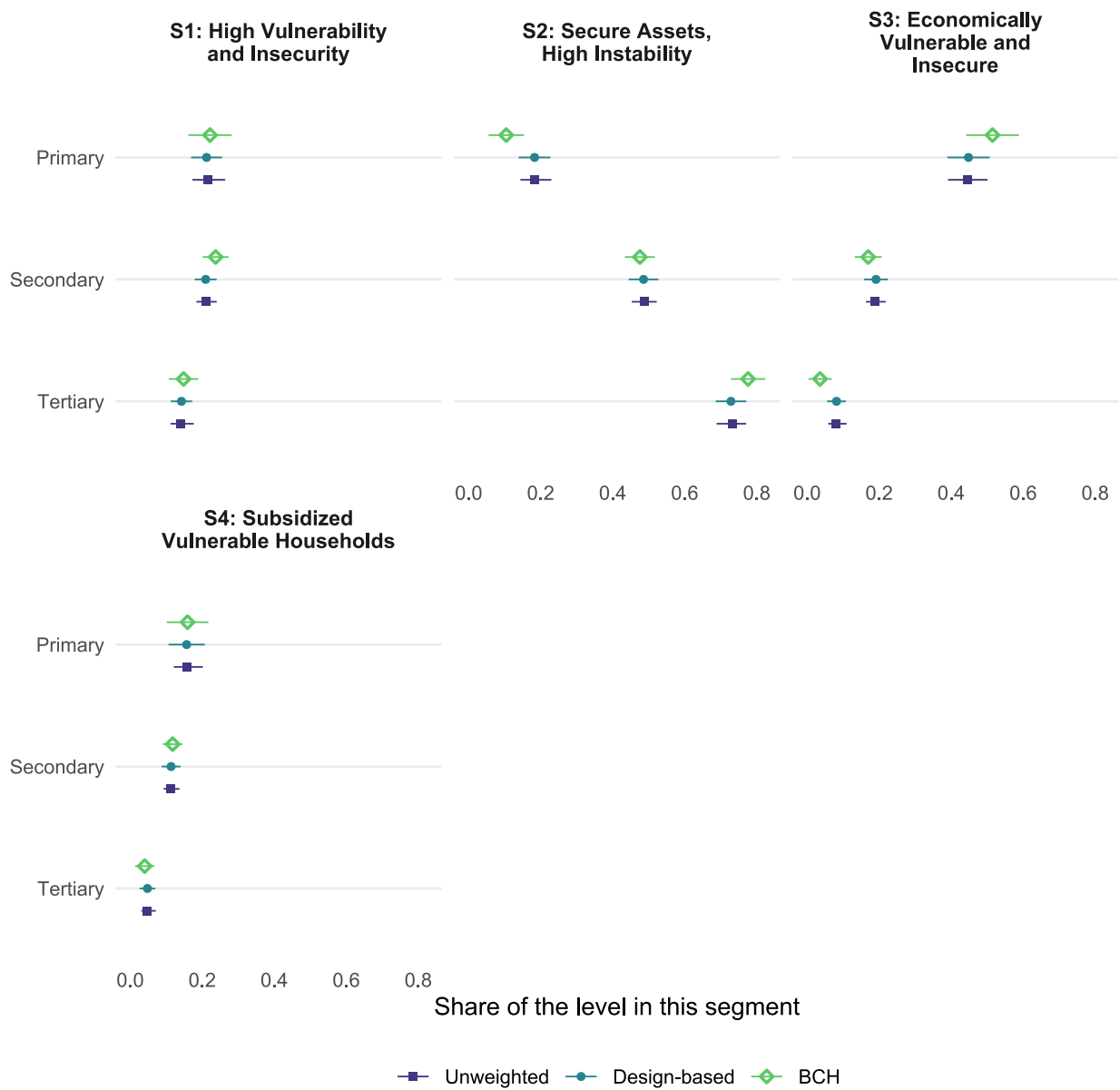
15.4 education

Table 16: Composition by education: design-weighted proportion among respondents who answered the item, with its JK_n standard error.

Level	n	Share
Secondary	798	0.513 (0.012)
Primary	312	0.201 (0.012)
Tertiary	447	0.285 (0.015)

Table 17: Share of each level of education falling in each segment under three estimators, with standard errors. Unweighted is a plain cross-tab with a binomial standard error. Design-based adds the weights and takes its variance from the JK_n replicates. BCH additionally removes the attenuation that modal assignment introduces. Columns sum to one down the segments. * fewer than 30 respondents; not a finding. † outside [0, 1], which the BCH correction permits and truncation would misstate.

Segment	Estimator	Primary	Secondary	Tertiary
S1: High Vulnerability and Insecurity	Unweighted	0.215 (0.023)	0.211 (0.014)	0.141 (0.016)
S1: High Vulnerability and Insecurity	Design-based	0.212 (0.022)	0.210 (0.015)	0.142 (0.015)
S1: High Vulnerability and Insecurity	BCH	0.222 (0.030)	0.237 (0.018)	0.148 (0.021)
S2: Secure Assets, High Instability	Unweighted	0.183 (0.022)	0.487 (0.018)	0.732 (0.021)
S2: Secure Assets, High Instability	Design-based	0.183 (0.022)	0.486 (0.021)	0.728 (0.021)
S2: Secure Assets, High Instability	BCH	0.104 (0.025)	0.475 (0.021)	0.776 (0.024)
S3: Economically Vulnerable and Insecure	Unweighted	0.446 (0.028)	0.189 (0.014)	0.081 (0.013)
S3: Economically Vulnerable and Insecure	Design-based	0.448 (0.030)	0.191 (0.017)	0.082 (0.013)
S3: Economically Vulnerable and Insecure	BCH	0.515 (0.037)	0.170 (0.019)	0.036 (0.016)
S4: Subsidized Vulnerable Households	Unweighted	0.157 (0.021)	0.113 (0.011)	0.047 (0.010)
S4: Subsidized Vulnerable Households	Design-based	0.157 (0.025)	0.114 (0.013)	0.048 (0.011)
S4: Subsidized Vulnerable Households	BCH	0.159 (0.029)	0.118 (0.014)	0.040 (0.013)



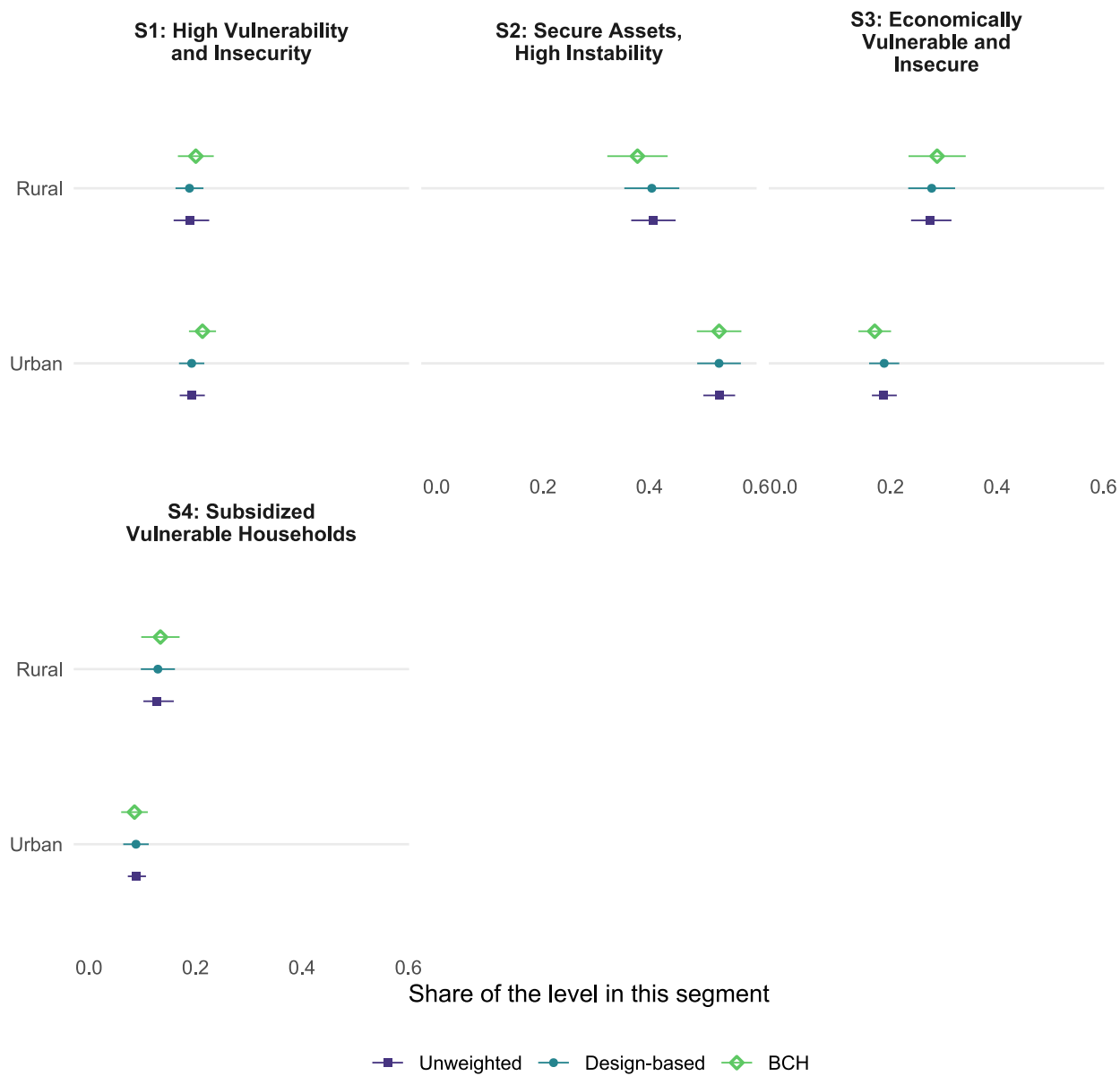
15.5 urban

Table 18: Composition by urban: design-weighted proportion among respondents who answered the item, with its JK_n standard error.

Level	n	Share
Urban	1073	0.659 (0.037)
Rural	531	0.341 (0.037)

Table 19: Share of each level of urban falling in each segment under three estimators, with standard errors. Unweighted is a plain cross-tab with a binomial standard error. Design-based adds the weights and takes its variance from the JK_n replicates. BCH additionally removes the attenuation that modal assignment introduces. Columns sum to one down the segments. * fewer than 30 respondents; not a finding. † outside [0, 1], which the BCH correction permits and truncation would misstate.

Segment	Estimator	Rural	Urban
S1: High Vulnerability and Insecurity	Unweighted	0.190 (0.017)	0.193 (0.012)
S1: High Vulnerability and Insecurity	Design-based	0.189 (0.013)	0.193 (0.012)
S1: High Vulnerability and Insecurity	BCH	0.201 (0.017)	0.213 (0.013)
S2: Secure Assets, High Instability	Unweighted	0.407 (0.021)	0.531 (0.015)
S2: Secure Assets, High Instability	Design-based	0.404 (0.026)	0.531 (0.021)
S2: Secure Assets, High Instability	BCH	0.378 (0.028)	0.531 (0.021)
S3: Economically Vulnerable and Insecure	Unweighted	0.275 (0.019)	0.187 (0.012)
S3: Economically Vulnerable and Insecure	Design-based	0.277 (0.022)	0.188 (0.014)
S3: Economically Vulnerable and Insecure	BCH	0.288 (0.027)	0.170 (0.015)
S4: Subsidized Vulnerable Households	Unweighted	0.128 (0.015)	0.089 (0.009)
S4: Subsidized Vulnerable Households	Design-based	0.129 (0.016)	0.088 (0.012)
S4: Subsidized Vulnerable Households	BCH	0.134 (0.018)	0.086 (0.013)



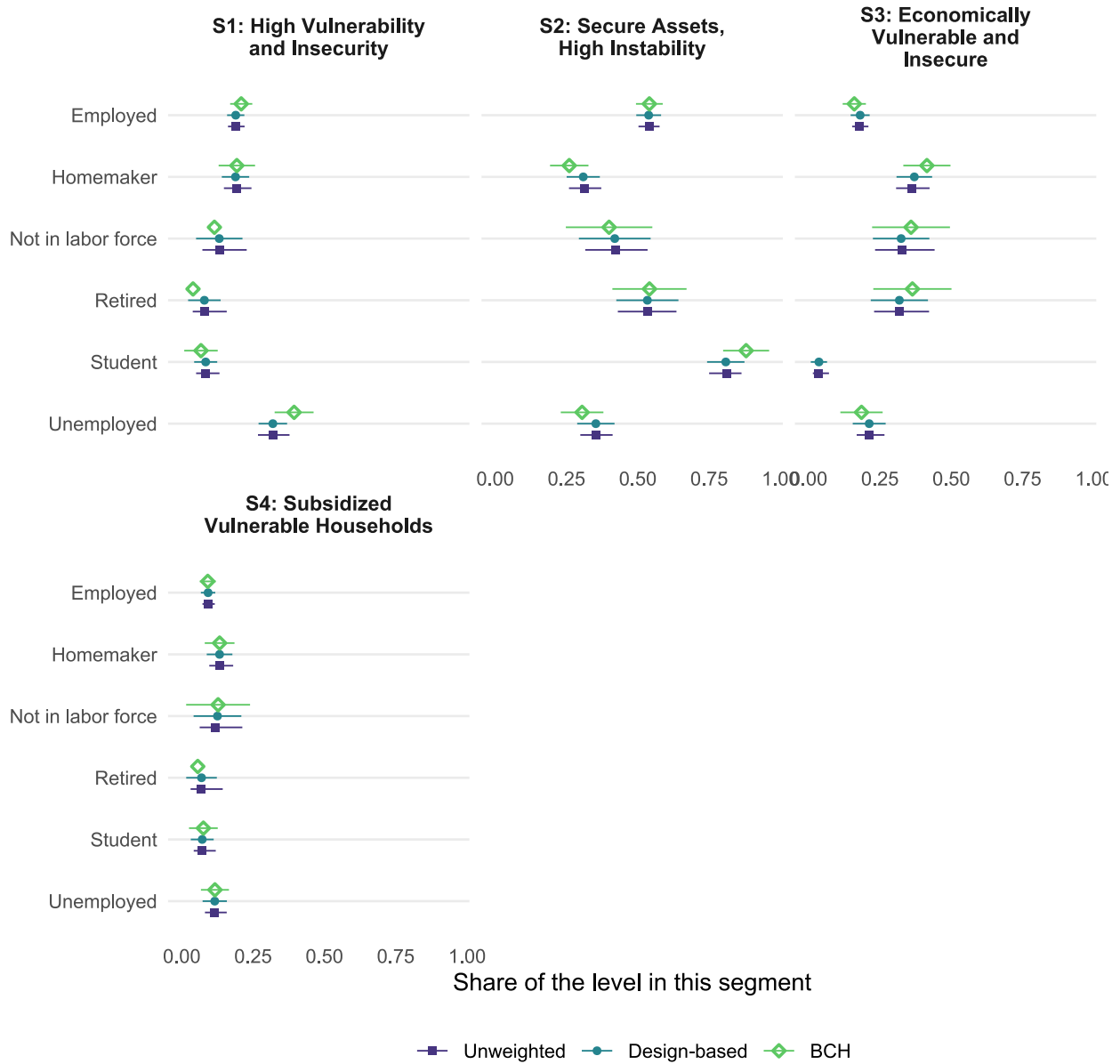
15.6 employment

Table 20: Composition by employment: design-weighted proportion among respondents who answered the item, with its JKn standard error.

Level	n	Share
Employed	697	0.443 (0.014)
Unemployed	272	0.173 (0.009)
Not in labor force	76	0.049 (0.006)
Student	182	0.115 (0.008)
Homemaker	256	0.164 (0.010)
Retired	88	0.056 (0.006)

Table 21: Share of each level of employment falling in each segment under three estimators, with standard errors. Unweighted is a plain cross-tab with a binomial standard error. Design-based adds the weights and takes its variance from the JKn replicates. BCH additionally removes the attenuation that modal assignment introduces. Columns sum to one down the segments. * fewer than 30 respondents; not a finding. † outside $[0, 1]$, which the BCH correction permits and truncation would misstate.

Segment	Estimator	Employed	Home-maker	Not in labor force	Retired	Student	Unemployed
S1: High Vulnerability and Insecurity	Un-weighted	0.189 (0.015)	0.191 (0.025)	0.132 (0.039)	0.080 (0.029)	0.082 (0.020)	0.320 (0.028)
S1: High Vulnerability and Insecurity	Design-based	0.189 (0.015)	0.188 (0.024)	0.131 (0.041)	0.079 (0.029)	0.084 (0.020)	0.319 (0.025)
S1: High Vulnerability and Insecurity	BCH	0.208 (0.019)	0.193 (0.032)	0.114 (0.059)	0.039 (0.043)	0.067 (0.029)	0.394 (0.034)
S2: Secure Assets, High Instability	Un-weighted	0.539 (0.019)	0.312 (0.029)	0.421 (0.057)	0.534 (0.053)	0.813 (0.029)	0.353 (0.029)
S2: Secure Assets, High Instability	Design-based	0.538 (0.022)	0.309 (0.029)	0.419 (0.063)	0.533 (0.055)	0.808 (0.033)	0.353 (0.033)
S2: Secure Assets, High Instability	BCH	0.540 (0.024)	0.260 (0.034)	0.399 (0.076)	0.541 (0.065)	0.880 (0.041)	0.305 (0.038)
S3: Economically Vulnerable and Insecure	Un-weighted	0.179 (0.015)	0.363 (0.030)	0.329 (0.054)	0.318 (0.050)	0.033 (0.013)	0.213 (0.025)
S3: Economically Vulnerable and Insecure	Design-based	0.181 (0.017)	0.371 (0.031)	0.324 (0.050)	0.318 (0.050)	0.036 (0.014)	0.212 (0.029)
S3: Economically Vulnerable and Insecure	BCH	0.160 (0.020)	0.415 (0.042)	0.359 (0.069)	0.364 (0.069)	-0.023 (0.020)†	0.185 (0.037)
S4: Subsidized Vulnerable Households	Un-weighted	0.092 (0.011)	0.133 (0.021)	0.118 (0.037)	0.068 (0.027)	0.071 (0.019)	0.114 (0.019)
S4: Subsidized Vulnerable Households	Design-based	0.092 (0.013)	0.132 (0.023)	0.125 (0.042)	0.069 (0.027)	0.072 (0.020)	0.116 (0.021)
S4: Subsidized Vulnerable Households	BCH	0.091 (0.014)	0.133 (0.026)	0.127 (0.056)	0.056 (0.036)	0.076 (0.025)	0.116 (0.025)



15.7 What the domain tables say

The tables above carry one row per segment for every demographic, and most of the cells say nothing. Finding the two or three that do is the analyst's problem, and this section drafts that reading with a language model.

The division of labour is the point. Before anything is sent, R works out which pairs of levels the design-based test resolves, and the model is handed that list and told to name only those. It never sees a standard error and never decides for itself whether a difference is real. This step goes to the editor rather than the worker: the arithmetic is settled before the call, but choosing which two or three of twenty-odd cells to point an analyst at still means holding every level of the demographic in view at once, and it is one call per demographic rather than one per segment. Its rules forbid ranking levels whose intervals overlap, forbid causal language, require silence on any level flagged as too small to characterise, and cap the output at four findings, because the point is to say where to look rather than to narrate every cell.

The prompt below is the real one, printed verbatim for the first demographic.

```
===== SYSTEM PROMPT =====
```

```
You are a senior survey methodologist who reads latent class analysis (LCA) measurement models. In this work each latent class is called a SEGMENT; that is a word-choice preference and the statistical object is unchanged. Each segment is described only by its item-response probabilities: for every survey item, the probability that a member of that segment gives each answer. A segment leans toward the answers with high probability. You interpret a segment strictly from these probabilities and the item wording, never from outside assumptions.
```

```
===== USER PROMPT ( age_cat ) =====
```

SURVEY CONTEXT

```
These items come from the 2023 AmericasBarometer survey of Ecuador, conducted by the LAPOP Lab at Vanderbilt University. Fieldwork was carried out face to face in Spanish by IPSOS between February and April 2023 with 1,604 respondents, drawn by a multi-stage probability design stratified by the three major regions of the country: Costa, Sierra, and Oriente.
```

```
The items analysed here record the material circumstances of the household and the respondent's own situation: whether the household owns particular durable goods, whether it has run short of food or water, whether finances and income have improved or worsened, whether the respondent feels unsafe or has been a victim of crime, whether the household receives government assistance or a conditional cash transfer, and whether the respondent intends to emigrate. The segments summarise patterns of economic vulnerability across these items.
```

COMPOSITION OF THE POPULATION AND OF EACH GROUP

```
age_cat in the population: 16-29 41% (n = 653), 30-44 26% (n = 420), 45-59 19% (n = 310), 60+ 14% (n = 221)
```

```
Share of each level falling in each segment:
```

High Vulnerability and Insecurity

16-29 0.17 [0.14, 0.20], 30-44 0.24 [0.20, 0.28], 45-59 0.20 [0.15, 0.24], 60+ 0.15 [0.10, 0.20]

Secure Assets, High Instability

16-29 0.59 [0.54, 0.64], 30-44 0.45 [0.39, 0.50], 45-59 0.44 [0.38, 0.51], 60+ 0.32 [0.25, 0.39]

Economically Vulnerable and Insecure

16-29 0.13 [0.10, 0.16], 30-44 0.21 [0.16, 0.25], 45-59 0.26 [0.21, 0.31], 60+ 0.46 [0.39, 0.52]

Subsidized Vulnerable Households

16-29 0.11 [0.08, 0.14], 30-44 0.10 [0.07, 0.14], 45-59 0.10 [0.07, 0.14], 60+ 0.07 [0.04, 0.11]

DIFFERENCES THE ANALYSIS RESOLVED

Criterion: pairwise design-based tests on the replicate covariance, Holm-adjusted within this demographic.

High Vulnerability and Insecurity: 30-44 differs from 60+

Secure Assets, High Instability: 16-29 differs from 60+

Secure Assets, High Instability: 16-29 differs from 45-59

Secure Assets, High Instability: 16-29 differs from 30-44

Economically Vulnerable and Insecure: 16-29 differs from 60+

Economically Vulnerable and Insecure: 30-44 differs from 60+

Economically Vulnerable and Insecure: 16-29 differs from 45-59

Economically Vulnerable and Insecure: 45-59 differs from 60+

Economically Vulnerable and Insecure: 16-29 differs from 30-44

WHAT THE SURVEY DESIGN CHANGES

High Vulnerability and Insecurity, 16-29: design moves the estimate by +0.000 and makes the interval 1.08 times as wide

High Vulnerability and Insecurity, 30-44: design moves the estimate by -0.002 and makes the interval 0.94 times as wide

High Vulnerability and Insecurity, 45-59: design moves the estimate by -0.001 and makes the interval 1.06 times as wide

High Vulnerability and Insecurity, 60+: design moves the estimate by -0.000 and makes the interval 1.02 times as wide

Secure Assets, High Instability, 16-29: design moves the estimate by -0.004 and makes the interval 1.30 times as wide

Secure Assets, High Instability, 30-44: design moves the estimate by -0.002 and makes the interval 1.19 times as wide

Secure Assets, High Instability, 45-59: design moves the estimate by -0.001 and makes the interval 1.11 times as wide

Secure Assets, High Instability, 60+: design moves the estimate by +0.001 and makes the interval 1.16 times as wide

Economically Vulnerable and Insecure, 16-29: design moves the estimate by +0.002 and makes the interval 1.18 times as wide

Economically Vulnerable and Insecure, 30-44: design moves the estimate by +0.004 and makes the interval 1.14 times as wide

Economically Vulnerable and Insecure, 45-59: design moves the estimate by +0.002 and makes the interval 1.05 times as wide

Economically Vulnerable and Insecure, 60+: design moves the estimate by -0.001 and makes the interval 0.98 times as wide


```
caption = "Drafted reading of each demographic. Verify  
against the tables above.")
```

Table 22: Drafted reading of each demographic. Verify against the tables above.

Variable	Finding	Caution
age_cat	The segments differ by age, with the 'Secure Assets, High Instability' segment having a higher share among 16-29 year olds than among older age groups, and the 'Economically Vulnerable and Insecure' segment having a higher share among those 60+ than among younger age groups, particularly 16-29 year olds. The data also separate 30-44 and 60+ in 'High Vulnerability and Insecurity', and 45-59 from 16-29 and 60+ in 'Economically Vulnerable and Insecure'.	Accounting for the survey design moved some estimates by up to 0.004 and changed the width of the confidence intervals by factors ranging from 0.94 to 1.30, with most intervals becoming wider.
sex	The segments 'Secure Assets, High Instability' and 'Economically Vulnerable and Insecure' differ between females and males. For 'Secure Assets, High Instability', 45% of females and 54% of males fall into this category, while for 'Economically Vulnerable and Insecure', the proportions are 26% of females and 18% of males. The data do not separate the other segments by sex.	Accounting for the survey design moved some estimates slightly, by up to 0.004, and made the intervals between 1.05 and 1.35 times as wide.
education	The segments differ by education level. For 'High Vulnerability and Insecurity', 'Economically Vulnerable and Insecure', and 'Secure Assets, High Instability', Primary and Secondary differ from Tertiary. For 'Secure Assets, High Instability' and 'Economically Vulnerable and Insecure', Primary also differs from Secondary.	Accounting for the survey design moved some estimates slightly, by up to 0.003, and changed interval widths by factors between 0.92 and 1.24.
urban	The segments 'Secure Assets, High Instability' and 'Economically Vulnerable and Insecure' differ between Rural and Urban areas. The data do not separate the other pairs. Look into what distinguishes these segments further.	Accounting for the survey design moved some estimates slightly (up to 0.002) and changed the width of the intervals by factors ranging from 0.79 to 1.40, with most intervals becoming wider.
employment	The segments show distinct patterns of economic vulnerability. The 'High Vulnerability and Insecurity' segment is more likely to include Unemployed respondents than any other employment category. The 'Secure Assets, High Instability' segment is dominated by Students, who differ from Homemakers, the Unemployed, and the Employed. The 'Economically Vulnerable and Insecure' segment is more likely to include Homemakers than Students or the Employed.	Accounting for the survey design moved some estimates slightly, by up to 0.007, and changed the width of the confidence intervals by factors ranging from 0.90 to 1.19, with most intervals becoming wider.

16 The delivered data product

The workflow writes a labelled SPSS file carrying the segment assignment with value labels, the label text, the posterior probabilities, and the assignment diagnostics, alongside CSV exports of the measurement model with intervals and the domain estimates.

The posterior columns are exported deliberately. A downstream analyst who cross-tabulates the modal assignment alone reintroduces exactly the attenuation described in Section 15, and the variable label on the segment column says so.

```
out_sav <- survey_scored |>  
  mutate(segment_label = if_else(is.na(segment), NA_character_,
```

```

        segment_labels$Label[segment]),
segment = haven::labelled(
  segment,
  labels = set_names(segment_labels$K, segment_labels$Label),
  label = paste("Segment (design-weighted LCA). Modal assignment:",
    "cross-tabs on this variable alone are attenuated by",
    "classification error. Use the post_segment",
    "columns for",
    "uncertainty-aware analysis.)))

haven::write_sav(out_sav, file.path(cfg$lca_dir, "survey_with_segments.sav"))

write_csv(mutate(ci_tbl, segment_label = segment_labels$Label[segment]),
  file.path(cfg$lca_dir, "measurement_model_with_CIs.csv"))
write_csv(transmute(domain_both, demographic = variable, level, segment,
  segment_label = segment_labels$Label[segment],
  estimator, prevalence = p, se, lo, hi),
  file.path(cfg$lca_dir, "segment_demographics.csv"))
write_csv(transmute(marginals, demographic = variable, level, n,
  weighted_share = weighted, se),
  file.path(cfg$lca_dir, "demographic_marginals.csv"))
write_csv(domain_reads, file.path(cfg$lca_dir, "domain_readings.csv"))

saveRDS(list(fit_w = fit_w, fit_unit = fit_unit, pl_aligned = pl_aligned,
  validation = validation, enum = enum, ci_tbl = ci_tbl,
  bvr_tbl = bvr_tbl, item_discrim = item_discrim,
  domain = domain,
  marginals = marginals, items = cfg$items,
  dat = dat, post_w = post_w, modal_int = modal_int,
  w_vec = w_vec, K = K_star, nu = nu),
  file.path(cfg$lca_dir, "model_fit.rds"))

glue("Wrote survey_with_segments.sav ({nrow(scoring)} scored), four CSVs, ",
  "and model_fit.rds to {cfg$lca_dir}")

```

Wrote survey_with_segments.sav (1604 scored), four CSVs, and model_fit.rds to D:/repos/weighted_inference_survey_estimation/output/lca

17 Limitations

The pseudo-likelihood can be multimodal with near-tied optima. The reported solution is the best mode found under many seeded starts, and the JK_n variance expresses within-mode design variance only, not uncertainty about mode selection.

The measurement model is fit on item-complete cases while scoring and domain estimation extend to partially observed respondents. Respondents scored from fewer items carry more classification error, which is the correlated-error case flagged in Section 2: a processing error whose magnitude depends on an item nonresponse error.

That split has a second consequence the classification-error argument does not cover: $\hat{\pi}$ and the domain estimates are not estimates for the same population. The prevalences in Section 13 describe item-complete responders; everything in Section 15 describes every respondent the model could place. If completeness is related to segment — and Section 14 argues it is — the two differ, and the gap is visible in this fit. Weighting the corrected domain estimates back up by the population composition returns segment shares of roughly 0.21, 0.48, 0.21 and 0.10 against the model’s 0.208, 0.500, 0.192 and 0.100: the correction recovers the model prevalence for the first and fourth segments and misses by about two points on the second and third. Some of that is rounding through the published tables and some is the frame. The clean fix is not a caveat. Section 6 restricts the $\hat{\rho}$ denominator to respondents who answered the item and a missing item contributes zero on the log scale, so the pseudo-likelihood already admits partial responders exactly; fitting on the scored frame rather than on complete cases removes the mismatch instead of documenting it, and it is inconsistent to assume ignorability for scoring while refusing it for estimation. This document still fits on complete cases, and that is the first thing to change in the next revision.

Domain estimates are reported both uncorrected and BCH-corrected. The uncorrected version uses hard modal assignment, which pulls composition toward the marginal distribution and so understates differences between demographic levels. It does not follow that a difference resolved on the uncorrected estimator is thereby safe: the correction moves the point difference and its standard error at the same time, the inverse classification table carries diagonal entries above one and negative entries off it, and the ratio of the two is not signed in general. Which of the two tests is the conservative one is an empirical property of this $\hat{D}$ and is not claimed here in either direction. The uncorrected test is the one reported because its intervals are the ones printed beside it.

The correction also assumes non-differential classification: one pooled $\hat{D}$ is inverted for every level of every demographic, which requires the classification error rates to be the same at each. That assumption is not tested, and Section 14 gives two reasons to doubt it holds exactly here — posterior sharpness falls with the number of items answered, and item nonresponse is not independent of age or education. The by-level assignment quality table in Section 14 is the diagnostic; where a level’s mean maximum posterior sits well below the others, its corrected row carries more error than its interval shows, and in a direction the interval cannot express.

The corrected version conditions on the fitted measurement model: step-one parameter uncertainty is not propagated in the reported intervals. This is standard three-step practice, and the sensitivity refit above shows what it costs. Rebuilding the measurement model inside every replicate multiplies the BCH standard errors for `age_cat` by a median of 2.17 and by as much as 4.22, against a weighted relative entropy of 0.71 rather than the high-separation regime in which fixed step one is usually defended [31]. Read the reported BCH intervals as optimistic by that median factor, and do not quote a separation that would not survive intervals of that width. These figures are computed by the sensitivity chunk above and are not carried over from an earlier render.

Bivariate residuals and the indicator screen are rankings without reference distributions. They order item pairs and items by how poorly the model accounts for them and support no test. The bivariate residual carries a further caution set out in Section 10: the value it takes under a correctly specified model rises with the size of the two-way table, so the ranking is not comparable across pairs with different numbers of categories, and the cell count is printed beside it for that reason.

The item discrimination D_j averages over segment pairs uniformly, so a pair of small segments counts as much as a pair of large ones. It is a property of the measurement model rather than of the population,

which is the intended reading, but it should not be described as the gap between two segments drawn at random unless segments are drawn uniformly rather than by prevalence.

The information criteria penalise on $\log n$, with n the number of sampled cases. Under clustering the effective sample size is smaller than n , so the penalty is weaker than a design-based criterion would impose and BIC leans toward more segments than the design supports. At the median design effect implied by the standard errors here the penalty would fall by about 0.18 per parameter, which narrows the gap between $K = 4$ and $K = 5$ from roughly 24 BIC units to 21 and changes nothing about the selection. On a design with more clustering it could.

The replicate count equals the number of PSUs, so the replicate covariance matrix has rank at most that number. Individual variances and pairwise covariances remain consistent, which is all the reported quantities require, but a joint Wald test across more dimensions than there are replicates is not available.

Where a design imposes quotas at the final selection stage, variables controlled by those quotas have suppressed between-cluster variance and standard errors that reflect fieldwork control rather than sampling information. Such variables' apparent precision should not be reported without comment.

Language model labels are drafts with an audited override path. No claim is made about their validity, and no reliability estimate across runs is reported.

18 Replication

`source_code.R` is the engine and is never edited per dataset. `survey_data_read.R` holds the file path, the design columns and the demographic recodes, and is shared with the factor report. `survey_lca_config.R` holds the item list and the settings for this arm. This document produces the analysis. A companion script audits object flow through the document in chunk order.

Six things change in a port.

The file path and item list. `item_codes` is a named character vector mapping analysis names to source variable names; the names become the analysis names automatically.

The nonresponse codes. `na_codes` lists the values representing nonresponse. The source file must be read with `user_na = TRUE`, since the default converts declared user-missing values to NA on import and the codes then never arrive.

The design variables. The configuration renames them to fixed names `id`, `strata`, `psu`, `wt`, so nothing downstream varies by dataset. Determining which source variables these are is the highest-consequence decision in a port: if the stratification variable is wrong, every standard error in this document is wrong and nothing else will reveal it.

The demographic recodes. One `recode_values` per variable, every observed source label named explicitly, with `unmatched = "error"` and an `NA ~ NA` arm. A regular expression scheme is not acceptable: a pattern that matches a label it was not intended to match silently misclassifies respondents, and one that matches nothing silently deletes them.

The profiling variable list. `cfg$aux` selects which demographics drive Section 15. Attitudinal variables should be excluded when they are plausibly correlated with the measurement items by construction, since profiling segments on them partly explains the measurement model with itself.

The configuration block. `K_range`, `n_starts`, `seed`, the information floor `min_items`, the complete-case switch, the output directory, and the language model endpoint.

The following halt the render rather than produce a wrong number: any demographic source label unmatched by its `recode_values` arms; any configured column absent from the data; any stratum containing a single PSU in the analysis frame, reported by stratum; a mismatch between the number of estimated domain quantities and their metadata; and a label file whose row count does not match K .

The following must be checked by the analyst: the recode audit, once per dataset; the weight coefficient of variation and unequal weighting effect, which determine whether weighting is doing anything in this file; the enumeration table and profiles, to choose K ; assignment quality by number of items answered, to set the information floor; and the drafted labels against their profile panels.

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